CPE334 Software Engineering Laboratory

TokTickIT Lab 4 Report

Actions Taken, final Ticket workflow and role dashboards

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Evidence checkpoint: 30 October 2026 (Asia/Bangkok)

REVIEW COPY - release, fresh main verification and final Done board pending

Accepted integration: b887994. Model and personal reflection confirmed. Release review, FROM-MAIN tests and final all-Done board are explicitly pending.

Contents

Answer Part 1 - Git Use with Engineering Workflow 3

Answer Part 2 - Spec DD 9

Answer Part 3 - Test DD and Traceability 14

Answer Part 4 - AI Use with Reflection 22

Answer Part 5 - Working IT Staff Dashboard UI 24

Answer Part 6 - Working Actions Taken UI 28

Answer Part 7 - Working Ticket Workflow 36

Answer Part 8 - Working Requester Dashboard and Final Regression UI 41

Answer Part 9 - Zen Green UI, Responsive, Accessibility, and Final Polish 45

Answer Part 1 - Git Use with Engineering Workflow

Repository: iceswift/toktickit. Author iceswift; accepted integration reviewer and merger Richyboy170. Phases 1-7 are accepted at lab4-staging b887994. Issue75 is Started on docs/lab4-release-report. This is an inspectable review copy, NOT the final submission: it does not claim a Lab4 release into main or an all-Done board.

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Lab 4 Peer Review Record

Author: iceswift. Integration reviewer/merger: Richyboy170. Snapshot30October2026, through accepted Phase7. Full actual metadata: artifacts/lab-04/phase8/review-records.json. Contemporaneous history preserves all original pending/fixing checkpoints; they are historical, not contradictory current states.

Issue / integration PR	Actual decision	Author response	Reviewer merge (UTC)
#68 / PR77	Richyboy170 approved final c0c9ba9; no unresolved inline finding recorded	Actual reply	Richyboy170, 0f84785, 2026-10-28 06:34:46
#69 / PR78	Richyboy170 approved final c2d7027; earlier findings corrected in same PR	Actual reply	Richyboy170, f4b6ea9, 2026-10-28 22:23:32
#70 / PR79	Richyboy170 approved final cc479b2; no unresolved inline finding recorded	Actual reply	Richyboy170, 2f633d0, 2026-10-29 11:36:31
#71 / PR80	Richyboy170 approved final 60c20d4; no unresolved inline finding recorded	Actual reply	Richyboy170, ae16e00, 2026-10-29 13:49:29
#72 / PR81	Richyboy170 approved final 6b12c82; no unresolved inline finding recorded	Actual reply	Richyboy170, aa1f868, 2026-10-29 14:53:40
#73 / PR82	Richyboy170 approved final d64d0e3; no unresolved inline finding recorded	Actual reply	Richyboy170, df785a5, 2026-10-30 05:01:08
#74 / PR83	Richyboy170 approved final bb8f60b; no unresolved inline finding recorded	Actual reply	Richyboy170, b887994, 2026-10-30 11:03:36

Substantive feedback and corrections

Original PR76: jarbbie requested changes to incorrect Requester HTTP prefixes. Correction fa8e550 uses /api/tickets and /api/attachments, preserves /staff,/admin,/auth, and adds a contract guard. Author reply. Student requested reviewer handover; corrected contract approved/merged via PR77. GitHub also marks76 merged as its commits are contained in77. No approval on76 is invented. Original direct Issue68 link is on76; replacement77 had no direct Development link at its original merge. This traceability gap is disclosed, not retrospectively described as timely compliance.

PR78 feedback: uniqueness of event versions alone did not make parent/entity/history versions atomic or sequential. Correction c031106 adds shared locked versioned mutations and14 real transaction cases, with integrated migration/reseed/recovery regressions. Both findings answered in the same discussion before final approval. No history deletion or weakened assertions used.

Verified workflow and release boundary

For accepted product work, actual reviewer approval precedes reviewer merge; author replies and same-PR fixes retained. Issue closure/Project Done is checked after acceptance, not inferred from opening a PR. PR83 reviewed head bb8f60b and merge b887994 are tree-identical; Phase7 closure records proof.

Phases1-7 are integrated into lab4-staging, not yet released Lab4 main. Phase8 report and release reviews follow phase8-release.md. Final report generation reads dated release facts from actual GitHub metadata and the fetched main SHA; this feature-integration record does not predict its own future release event. No author merge, fabricated review dialogue or CI Pass is claimed.

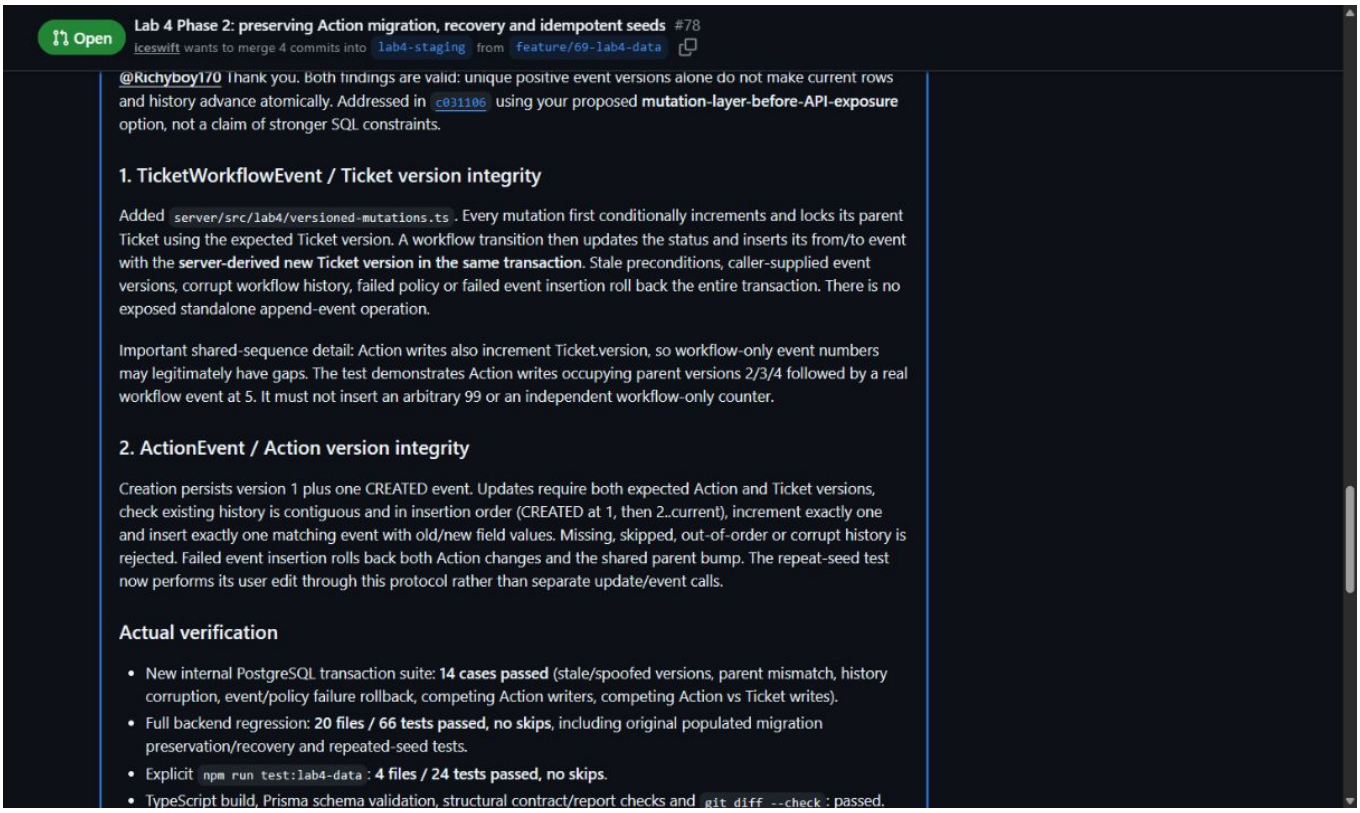


Figure 1. Actual substantive version/history feedback and author correction in PR78; not a fabricated review conversation. The linked discussion retains full text. Full original

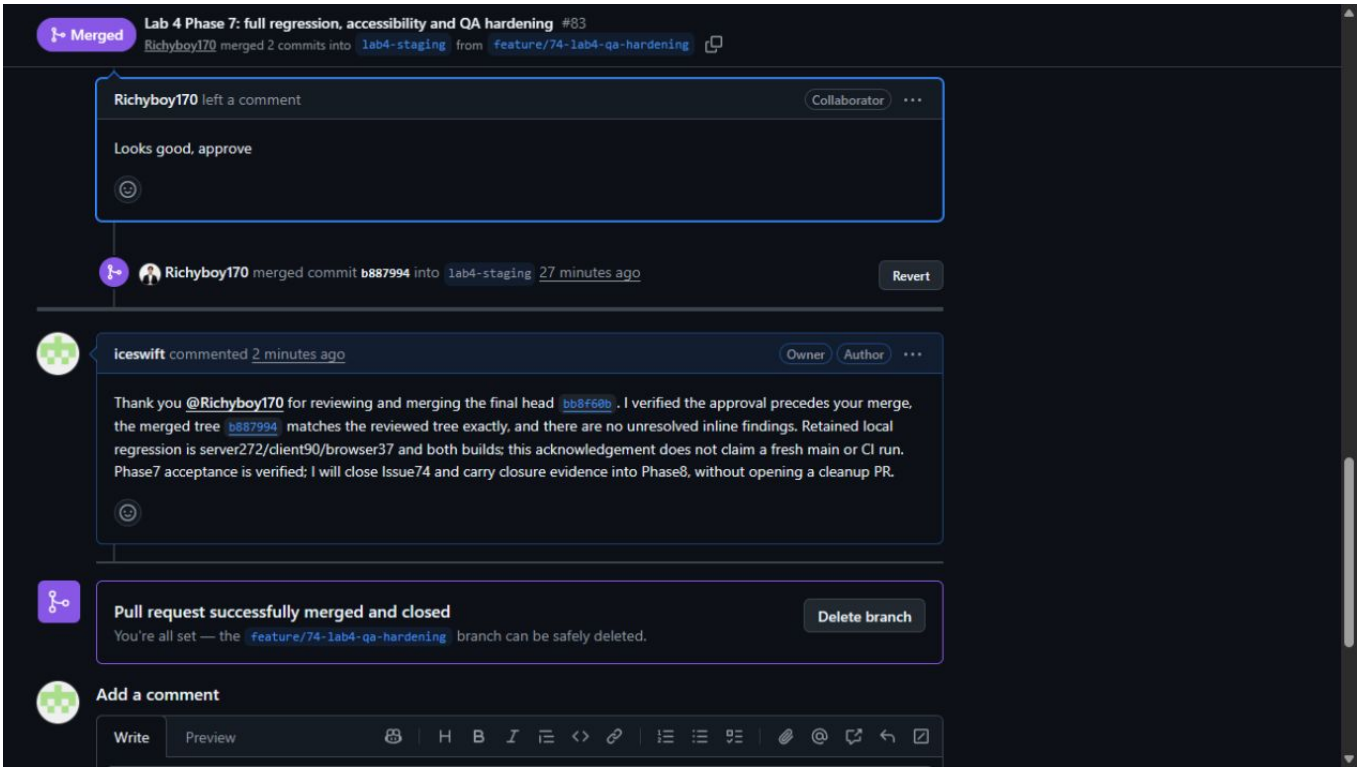


Figure 2. Latest accepted feature PR83: Richyboy170 approved and merged; author acknowledgement linked above. Full original

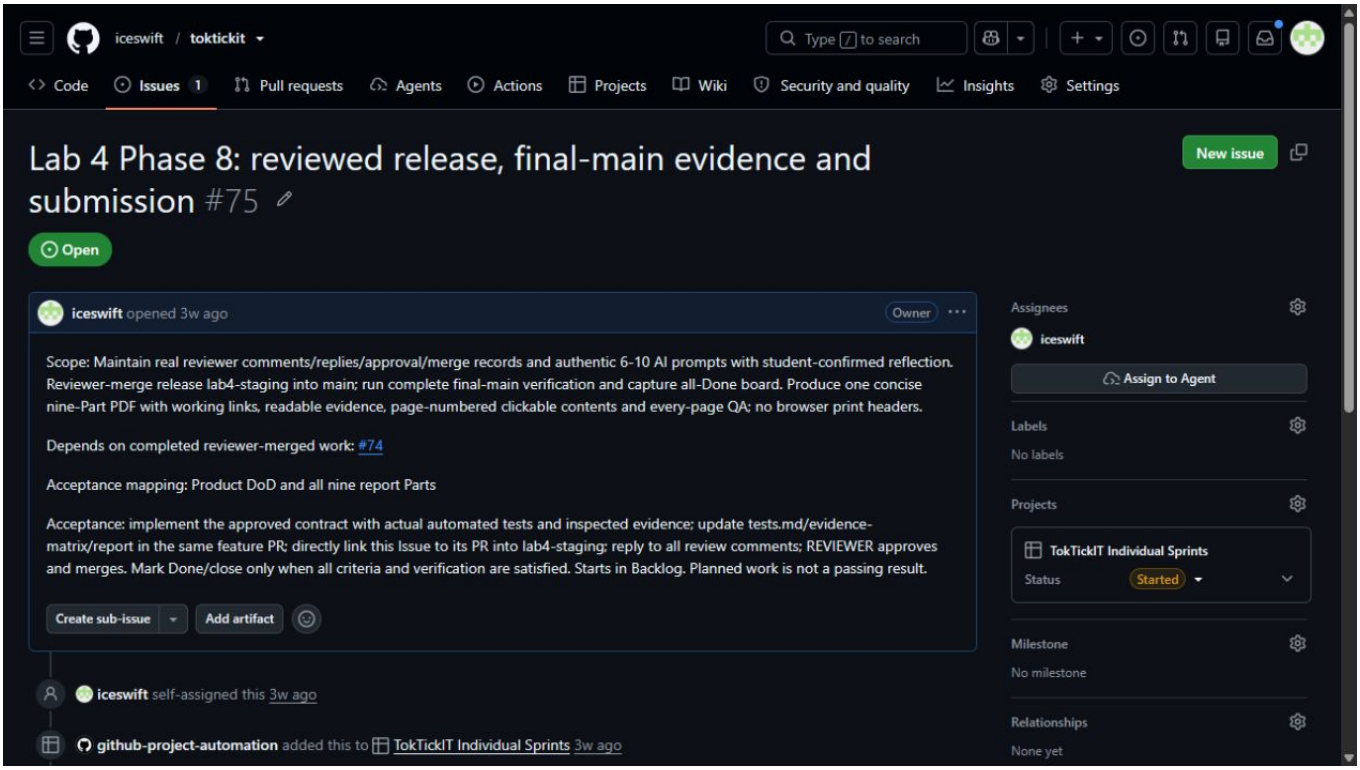


Figure 3. Current checkpoint, NOT final Kanban: report/release Issue75 remains Started. A final all-eight-Issue Done capture is required only after all release gates hold. Full original

Actual feature-to-staging graph, excluding released Lab3 baseline5398f2d. Lab4 staging-to-main release has not happened at this checkpoint.

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```
*   b887994 Merge pull request #83 from iceswift/feature/74-lab4-qa-hardening
|\
| * bb8f60b docs(lab4): record Phase7 linked peer-review handoff
| * 174bd18 feat(lab4): complete Phase7 regression and accessibility hardening
|/
*   df785a5 Merge pull request #82 from iceswift/feature/73-lab4-dashboards
```

```
|\
| * d64d0e3 docs(lab4): record Phase6 PR82 direct linkage and review evidence
| * edb2c2a feat(lab4): add authoritative role dashboards and verified drill-down
|/
*   aa1f868 Merge pull request #81 from iceswift/feature/72-lab4-ticket-workflow
|\
| * 6b12c82 docs(lab4): retain Phase 5 linked PR and peer-review checkpoint
| * 213df6e feat(lab4): enforce final Ticket workflow with audited gate and UI recovery
|/
*   ae16e00 Merge pull request #80 from iceswift/feature/71-lab4-actions-ui
|\
| * 60c20d4 docs(lab4): record Phase 4 linked PR and inspected review checkpoint
| * 4820ce3 feat(lab4): implement role-safe Actions Taken UI with verified evidence
|/
*   2f633d0 Merge pull request #79 from iceswift/feature/70-lab4-actions
|\
| * cc479b2 docs(lab4): verify linked ready Phase3 PR and actual review state
| * bfd2089 feat(lab4): add authorized versioned Action APIs and durable retries
|/
*   f4b6ea9 Merge pull request #78 from iceswift/feature/69-lab4-data
|\
| * c2d7027 docs(lab4): retain reviewer reply and actual correction proof
| * c031106 fix(lab4): enforce atomic versioned history before API exposure
| * e17403f docs(lab4): verify linked ready PR and in-app Kanban evidence
| * 2192131 docs(lab4): record phase2 draft submission and normalize log whitespace
| * 0ad56c4 feat(lab4): add preserving Action data migration and repeat-safe seeds
|/
* 0f84785 Merge pull request #77 from iceswift/docs/lab4-contract-richy
* c0c9ba9 docs(lab4): record student-requested reviewer handover
* 9e24106 docs(lab4): retain actual route review response evidence
* fa8e550 docs(lab4): resolve requester HTTP paths from peer review
* fc1cdfa docs(lab4): record linked phase1 PR and inspected review evidence
* 753ddeb docs(lab4): establish approved engineering contract before implementation
```

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TokTickIT

React/TypeScript/Bootstrap frontend, Express4 API and Prisma5/PostgreSQL service desk. Labs1-3 provide Tickets, authenticated Requesters/Staff/Admin, ownership, Attachments, Public Comments, private Internal Notes and account administration. Lab4 adds audited Actions Taken, safe final Ticket workflow and role dashboards.

Current delivery boundary

Lab4 Phases1-7 are independently approved and merged into lab4-staging. Phase7 PR83 was approved/merged by Richyboy170; Issue74 accepted/Closed/Done. Phase8 report/release is on docs/lab4-release-report (Issue75), not yet a reviewer-approved release. Phase8 will publish reviewed staging into main and verify the final single PDF/main output. No final-main/CI pass is inferred.

Setup (working development environment)

Use Node22.12+ (tested22.17), npm, Docker Desktop or PostgreSQL. With older npm10, Vitest optional-peer resolution may fail; a tested one-off resolver is npx --yes --package npm@11.6.2 npm ci, not a global Node/npm replacement.

1. docker compose up -d postgres. Existing compose maps PostgreSQL to host5434. 2. In server, copy .env.example to .env; configure your development DATABASE_URL. Run npm ci, npx prisma generate, npx prisma migrate deploy, npm run prisma:seed, then npm run dev (default API3000). 3. In client, copy .env.example to .env; configure VITE_API_URL if needed. Run npm ci and npm run dev; open localhost5173. 4. npm run build in server and client produces production builds. Production auth cookies require HTTPS; configure environment/origin deliberately.

Migrations are additive. Never reset a populated database to demonstrate Lab4. See populated migration/backup/recovery runbook.

Synthetic demonstration data

Seed is idempotent: existing passwords, worked Ticket fields, Actions/history and versions are preserved. It maintains categories/systems and legacy Users/Tickets, plus eight dedicated Lab4 examples spanning statuses/priorities/ownership and zero/one/multiple Actions. Unknown legacy resolution times stay null, not backfilled.

Demo accounts (example.test, local fixtures only):

Role	Email
Requester	amina.rahman@example.test
Requester	ben.carter@example.test
Staff	iris.nattapong@example.test
Staff	jonas.miller@example.test
Administrator	narin.admin@example.test

Initial fixture password Lab3Initial!2026; first login requires a real password change. Do not expose these fixtures publicly or reuse this credential elsewhere.

Main capabilities / HTTP paths

- GET /api/health, GET /api/categories; development requester enumeration retired.

- POST /auth/login, POST /auth/change-password, POST /auth/logout, GET /auth/me.

- POST/GET /api/tickets, GET /api/tickets/:id: session-owned Requester data.

- /api/tickets/:id/attachments, /api/attachments/:id/download, soft removal.

- GET /api/tickets/dashboard: own Open/Waiting/recently-resolved, attention/recent.

- GET /staff/dashboard: operational status/priority/owner and performedBy metrics.

- /staff/tickets, /staff/tickets/:id: queue/detail/priority/owner operations.

- /staff/tickets/:id/actions: create/read/update/assign/status with explicit shared Ticket/Action versions. Immutable performer, append-only audit, persistent scoped Action-create idempotency. My Actions /staff/actions?performedBy=me is not assignee work.

- GET /staff/tickets/:id/workflow-events and PATCH /staff/tickets/:id/status: final transitions/gates/version conflict; Requester indication remains advisory. Staff and Administrator formal status permission is a Lab4 extension; Admin does not gain Staff-only owner/comment/Note writes.

- Requester own-Ticket Actions/workflow history are read-only; cross-owner access404.

- Public Comments shared with Requester; Internal Notes private to Staff/Admin.

- /admin/users: Administrator-only account management and safety constraints.

Exact methods/payloads/statuses: API contract. Dashboard is an explicit navigation choice; legacy role landing pages remain. Card destinations retain their calculation filters. Refresh gets a new snapshot.

Tests: never use the working database

Client npm test runs components/adapters; npm run build typechecks and builds. Backend tests include mutating fixtures; use a fresh reserved database on the dedicated test PostgreSQL cluster at127.0.0.1:55484, not Docker5434 or server/.env. No destructive reset/drop is required. Retain audit/history fixtures after runs.

Prepare fresh lab4_p7_reg_<runid> and lab4_p7_ui_<runid> using createdb, migrate deploy and seed with DATABASE_URL explicitly targeting each test DB. The cluster must permit creating additional isolated migration/recovery/perf DBs.

PowerShell, from server (replace runid/tool path with your actual isolated setup):

```
$env:DATABASE_URL='postgresql://postgres@127.0.0.1:55484/lab4_p7_reg_runid'
$env:LAB4_TEST_ADMIN_URL='postgresql://postgres@127.0.0.1:55484/postgres'
$env:LAB4_PG_BIN='C:/Program Files/PostgreSQL/18/bin'
$env:LAB4_PERF_TEST='1'
npm run test:lab4-qa
npm run build
```

From client:

```
$env:LAB4_UI_TEST='1'
$env:LAB4_UI_DATABASE_URL='postgresql://postgres@127.0.0.1:55484/lab4_p7_ui_runid'
npm run test:e2e
```

Browser QA config refuses working DB names/existing server reuse; launches isolated API55485/UI55486. Shared fixture helpers also validate opt-in before mutations. Run all preserved Lab2/3 scenarios and Lab4, not only the newest tests. Actual screenshots/axe reports go to output/playwright/phase7 for inspection before copying to artifacts. Missing DB/performance opt-in is not a passing result.

Performance smoke uses its own new10,000-Ticket/20,000-Action database, five warmups then20 HTTP requests per role, provisional p95<=1000ms. Logs include hardware, bounded ORM operation counts, response size and EXPLAIN ANALYZE. This is a local development smoke threshold, not production load/SLA certification.

Audit: npm audit --json in each package. Phase7 updates Vitest4.1.12/Vite6.4.5 and compatible server/transitive patches; never use audit-fix --force blindly. Real before/after results and remaining limits: QA record.

Repository / workflow / evidence

```
client/src/                 role UI and safe API adapters
client/tests/lab-01..04/     component/style/adapter tests
client/e2e/lab-02..04/       preserved and incremental browser acceptance
server/src/lab4/            transactional Actions/workflow/dashboard domain
server/prisma/              additive migrations and idempotent seeds
server/tests/lab-01..04/     unit/API/migration/recovery/performance tests
docs/lab-04/                contract, test traceability, review, AI-use, working report
artifacts/lab-04/           inspected captures, actual output and provenance
```

Issues and Kanban: feature branch -> directly Issue-linked PR into lab4-staging -> reviewer approves AND merges. Author replies to every review comment, fixes stay in same PR. Done only after actual acceptance and reviewer merge. Final reviewed Release PR into main is Phase8.

Real .env, credentials, uploads, dependencies, build output and private fixture DB data must never be committed; see .gitignore. No direct staging/main development.

Rendered .gitignore

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```
# dependencies
node_modules/
# env & secrets
.env
*.env
!.env.example
# build output
dist/
build/
client/test-results/
client/playwright-report/
# prisma
server/prisma/*.db
server/uploads/
# logs & OS
*.log
.playwright-cli/
output/
.DS_Store
```

Answer Part 2 - Spec DD

Original engineering contract753ddeb predates implementation. Corrected Requester HTTP prefixes fa8e550 were reviewed before PR77 integration. The current rendering below consolidates the approved FR/BR/AC, role and status matrices, query predicates, migration decisions and truthful DoD. Student-approved design choices are distinguished from instructor requirements.

Rendered docs/lab-04/specification.md

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Sprint 4 Engineering Contract

Original contract753ddeb (8October2026) predates product implementation; D-01..04 were student-approved. Route correction fa8e550 was peer reviewed before contract integration PR77/0f84785 (28October). Baseline released Lab3:5398f2d. This current rendering consolidates the SAME approved rules, not a new product scope. Earlier checkpoints and Git history preserve provenance. Phases1-7 are reviewer-merged at b887994; final release/main/PDF gates remain open.

1. Sprint goal and stakeholder interpretation

Complete the service-desk workflow with traceable Actions Taken, authoritative role dashboards and safely enforced resolution, preserving all earlier behavior. The Ticket Owner coordinates the Ticket; other permitted staff can record work.

2. Scope

Include Actions list/create/assign/edit/complete/cancel, final Ticket workflow, Requester and Staff/Admin dashboards, populated-data migration/recovery, idempotent seeds, conflict/retry protection and full regression/accessibility. Exclude notifications, SLA/escalation, inventory/cost/payroll, signatures, advanced BI, multitenancy and unapproved features.

3. Student-approved decisions

ID	Student-approved decision	Why an explicit design decision was needed
D-01	Authenticated creator/performedBy is immutable and separate from active Staff/Admin assignee; another staff member may record work. Requester reads all Actions on owned Tickets only.	Assignment is graded but the handout does not fully specify actor vs assignee.
D-02	PLANNED -> IN_PROGRESS -> COMPLETED; PLANNED/IN_PROGRESS -> CANCELLED. Terminal records stay visible; no deletion/reopening.	Action states/transitions are graded but exact values are not supplied.
D-03	Editable current Action with immutable append-only revision/assignment/status events, sorted by timestamp then ID.	Required editing must coexist with append-only evidence.
D-04	Resolve requires >=1 COMPLETED Action, no PLANNED/IN_PROGRESS Actions, and no outstanding follow-up. Cancelled work does not satisfy completion.	The handout requires a gate without prescribing its complete predicate.

D-04 preserves legacy Resolved/Closed records unchanged; zero-Action legacy Tickets remain readable, but a new Resolve transition must pass the same gate. These choices are student-approved, not instructor-prescribed values. Exact details below are engineering choices under this scope for the peer reviewer to check.

4. Numbered requirements

ID	Requirement	Acceptance criteria
FR-01	Staff/Admin manage Actions Taken under exactly one Ticket with date/time, description, result, auto actor, follow-up flag/note and attachment notes.	AC-01,02,03
FR-02	Requesters read all owned-Ticket Actions but cannot write or read another Requester's Ticket.	AC-04
FR-03	Enforce final Ticket transitions and resolution gate in backend; Requester resolution indication stays advisory.	AC-05,06
FR-04	Backend computes Requester-owned metrics/recent/attention items with matching drill-down.	AC-07
FR-05	Backend computes Staff/Admin operational metrics, current-user Actions and recent/urgent items.	AC-08
FR-06	Preserve previous records/relationships and regressions; idempotent demonstration seeds.	AC-09,10
FR-07	Detect stale writes, prevent duplicate retries and retain input after recoverable failures.	AC-11,12
FR-08	Coherent Zen Green navigation and usable desktop/tablet/mobile and keyboard interaction.	AC-13

5. Business rules

- BR-01: An Action belongs to exactly one existing Ticket.

- BR-02: Ticket Owner coordination is separate from performed-by work.

- BR-03: All writes require backend authentication and role authorization.

- BR-04: Follow-up Note is required whenever follow-up is required.

- BR-05: Requester indication does not set Ticket status to Resolved.

- BR-06: Preserve all eight Ticket statuses, legacy records and private Notes.

- BR-07: Dashboard calculations use authoritative backend data and documented ranges; Requester identity comes from session, never client-supplied identity.

- BR-08 (D-01): Reject inactive/wrong-role assignees and spoofed actor.

- BR-09 (D-02): Enforce explicit Action transitions server-side.

- BR-10 (D-03): Events cannot be overwritten/deleted through application APIs.

- BR-11 (D-04): Apply the resolution predicate even to direct API calls.

- BR-12: Trim text: Description 1-2000 UTF-16 code units; Result 0-2000, required/nonempty at completion; Follow-up Note 0-2000, required/nonempty when flag true; Attachment Notes 0-2000. Escape plain text, never interpret HTML.

- BR-13: Required actionDateTime is timezone-qualified ISO 8601, between Ticket createdAt and server validation time inclusive. Immutable server createdAt is separate.

- BR-14: New Actions start PLANNED with an active Staff/Admin assignee. Fields cannot spoof/change Ticket, creator, creation time or status through a general edit.

- BR-15: Staff/Admin may record/update work on any accessible active Ticket, not only their own assignment. RESOLVED/CLOSED/CANCELLED Tickets' Actions are read-only.

- BR-16: Terminal Actions cannot transition again; eligible field revisions on active Tickets append history. COMPLETED Result remains nonempty. Clearing follow-up is an audited edit; old note/flag are retained in history.

- BR-17: Ticket cancellation requires no PLANNED/IN_PROGRESS Actions; cancel these explicitly first. CLOSED/RESOLVED may reopen; CANCELLED cannot reopen.

- BR-18: Every Action mutation increments parent Ticket version/updatedAt atomically. Resolve and Action creation/follow-up edits share a parent-version transaction, preventing a race that bypasses the resolution gate.

- BR-19: Scoped creation retry returns the original Action without another event; reused key with different normalized payload returns 409.

6. Preserved Ticket matrix and final role/gate rules

Verified in baseline server/src/app.ts:

From	Existing destinations
NEW	OPEN, IN_PROGRESS, CANCELLED
OPEN	IN_PROGRESS, WAITING_FOR_REQUESTER, RESOLVED, CANCELLED
REOPENED	IN_PROGRESS, WAITING_FOR_REQUESTER, RESOLVED, CANCELLED
IN_PROGRESS	WAITING_FOR_REQUESTER, RESOLVED, CANCELLED
WAITING_FOR_REQUESTER	IN_PROGRESS, RESOLVED, CANCELLED
RESOLVED	CLOSED, REOPENED
CLOSED	REOPENED
CANCELLED	none

Final destinations preserve the baseline matrix above, adding BR-11/17/18 gates. All unlisted transitions return 409. Administrator status writes are an explicit Lab 4 extension of the baseline IT-Staff-only permission, not a Lab 3 claim.

Operation	Requester	Staff	Administrator
Read all Actions/work revisions	Owned Tickets only	All Tickets	All Tickets
Create/edit/assign/transition Action	Never	Active Tickets	Active Tickets
Formal Ticket transition	Never	Matrix+gate/version	Matrix+gate/version
Advisory indication	Preserved owned-Ticket behavior	No impersonation	No impersonation
Dashboard	Owned data	Operational data	Operational data
InternalNotes/owner/comments	Preserve baseline permissions	Preserve baseline	Preserve baseline

Shared Action revisions never include private InternalNotes/session data. Inactive/must-change Users retain baseline denial behavior.

7. Data design and preservation

At the released Lab3 baseline, models were Category, RelatedSystem, DevelopmentRequester, User, AuthSession, Ticket, Attachment, PublicComment and InternalNote. Seven migrations existed. Lab4 adds the eighth migration and the work/version/history/dashboard increment.

1. Add Action plus immutable history rather than repurposing PublicComments or InternalNotes: work results and revision provenance are distinct concepts. 2. Versioned transactional writes prevent silent stale overwrites (409); history and current record updates must commit atomically. 3. Index Action by Ticket and deterministic date/ID, and assignee/status for dashboards. Validate query plans with a documented performance fixture. 4. Scoped unique idempotency keys prevent duplicate create on retries; payload mismatch returns 409. No reliance solely on disabled submit buttons.

Test migration on an isolated populated legacy database with before/after IDs, counts and relationships, then test documented backup/restore recovery. Do not run destructive demonstrations against the user's working database. Seeds must cover every major Ticket state/priority/ownership, zero/one/multiple Actions and zero/non-zero metrics; repeat runs must not duplicate records.

8. Exact metrics and final data choices

Store timestamps in UTC; use Asia/Bangkok calendar boundaries converted to UTC, half-open [start,end) ranges. One response has one asOf instant. Open means NEW/OPEN/IN_PROGRESS/WAITING_FOR_REQUESTER/REOPENED. Recent ordering: updatedAt DESC, id DESC, limit 5; do not call this a time-window count. Requester: open, waiting for requester, recently resolved and recent updates. Staff: unassigned open, current-owner open, status/priority groups, current-user Actions (performedBy, separate from assigned work), recent/urgent Tickets. Every card needs its exact query, zero state and matching drill-down filter.

One consistent read transaction supplies all metrics and lists. Seven-calendar-day window starts Bangkok midnight six days before response date and ends at asOf (exclusive). Ticket gains nullable lastResolvedAt set on genuine resolution; legacy Resolved/Closed dates remain unknown/null, excluded from time-window metrics but included in ordinary status counts. Do not invent historical events or timestamps.

Metric/list	Exact predicate and drill-down
Requester openTickets	session requesterUserId AND open statuses; GET /api/tickets?statusGroup=open
Requester waitingTickets	same owner AND WAITING_FOR_REQUESTER; status drill-down
Requester recentlyResolvedTickets	same owner AND RESOLVED/CLOSED AND lastResolvedAt in [start,asOf); resolvedSince/resolvedBefore and statusGroup=resolved
Requester recentTickets	same owner, all statuses, updatedAt DESC,id DESC, limit 5; detail links
Requester attentionTickets	same owner, WAITING_FOR_REQUESTER, same order/limit; detail links
Staff unassignedOpenTickets	open AND ownerUserId=null; statusGroup=open&unassigned=true
Staff myOpenTickets	open AND ownerUserId=session.id; open+ownerId drill-down
Staff ticketsByStatus	all Tickets grouped by each enum value incl. zero; status drill-down
Staff openTicketsByPriority	open grouped by every IT Priority incl. zero; open+priority drill-down
Staff myActions	performedByUserId=session.id, all states/all time; /staff/actions?performedBy=me
Staff recentTickets	all Tickets, updatedAt DESC,id DESC, limit 5; staff detail links
Staff urgentTickets	open AND HIGH/URGENT; URGENT first then updatedAt DESC,id DESC, limit 5

All drill-downs use identical predicates. Use actual HTTP paths relative to the server origin as defined in api-spec.md: Requester list/detail/actions use /api/tickets, while Staff paths remain /staff (no /api). These endpoint paths are not frontend navigation route names. Staff/Admin current-user metrics use actual session identity, not assignee. Recently-resolved records may include subsequently Closed work, never Reopened/Open work. Empty counts/buckets are zero, lists [].

Data increment: ActionTaken with fields listed in API plus Ticket/actor/assignee foreign keys, status, version default 1 and timestamps; ActionEvent immutable type/actor/time/version/changed-fields; TicketWorkflowEvent from/to/actor/time/version; IdempotencyRecord unique user+Ticket+key with normalized payload hash and Action ID. Ticket adds version default 1 and nullable lastResolvedAt. Preserve all nine tables and legacy records. No hard-delete/cascade-purge interface for work/history. Index Action(ticketId,actionDateTime,id),

(performedByUserId,createdAt), (assigneeUserId,status); events(ticketId,createdAt,id); dashboard composite indexes on requester/status and status/priority after validating query plans.

9. Observable acceptance criteria

- AC-01: Valid Action persists under correct Ticket with session actor and permitted assignee.

- AC-02: Assignment/edit/transitions/complete/cancel follow the approved matrix.

- AC-03: Required/conditional fields, dates and inactive assignees are validated.

- AC-04: Requester sees all own-Ticket Actions, no private Notes, cannot mutate or cross ownership.

- AC-05: All allowed Ticket transitions succeed and every other transition is rejected.

- AC-06: Resolution bypass fails without satisfying approved work/follow-up gate; indication is advisory.

- AC-07: Requester counts and drill-down agree with owned database fixture queries.

- AC-08: Staff/Admin metrics, including current-user Actions, agree with authoritative queries.

- AC-09: Populated legacy migration preserves records and tested recovery restores them.

- AC-10: Repeated seeds preserve demonstration data without duplicates.

- AC-11: Stale/concurrent writes return conflict without overwriting newer state.

- AC-12: Duplicate retry is safe; loading/validation/empty/forbidden/not-found/failure retain appropriate state.

- AC-13: All major screens pass defined responsive, semantic, keyboard and visual checks.

10. Product Definition of Done

- [x] Student-approved decisions, FR/BR/AC and exact role/transition/metric choices before code.

- [x] Corrected contract peer-approved/merged before implementation (PR77).

- [x] All ACs mapped to real unit/API/UI/style/auth/workflow/migration/E2E files (tests.md).

- [x] Populated migration, backup/restore recovery and repeated seeds verified.

- [x] Phase7 local backend272/client90/browser37, builds, dependency audit and performance pass.

- [x] Version/role/ownership/retry/concurrency and inspected three-width evidence retained.

- [x] Feature integrations through83 reviewer-approved/merged; comments answered.

- [x] Current README, ten authentic selected prompts, student-confirmed Lab4 reflection.

- [ ] Phase8 report changes independently reviewed/integrated and release merged into main.

- [ ] Complete fresh passing output from released main with exact source/environment.

- [ ] Final all-sprint-Issues Done screenshot and release observation verified.

- [ ] One concise PDF, nine ordered Parts, working links, page-numbered clickable contents, readable evidence, every-page render QA and no browser print headers.

UI controls/modes/feedback are defined in ui-spec.md; exact HTTP paths/payloads/safe status codes in api-spec.md. Data recovery: database-runbook.md. Implemented coverage: tests.md. Local feature verification is not silently relabelled final main/CI evidence.

Answer Part 3 - Test DD and Traceability

Accepted Phase7 local results: backend272/29files, client90/19files, browser37 scenarios, both builds and dependency audits passed. These are feature-tree results, NOT fresh main or CI results. Phase8 reruns cover the small favicon correction and preserved browser regressions. Complete FROM-MAIN output remains a hard final-submission gate after reviewer release.

Rendered docs/lab-04/tests.md

Exact source / complete raw output

Lab 4 Test DD and traceability - accepted feature verification

30 October 2026. Current matrix supersedes earlier planning-only tables. Historical red/green results and closures remain in phase2..6 records and Git. Base: reviewer-merged staging df785a5. Results are LOCAL feature-tree runs, not final main or CI. Phase7 was subsequently approved/merged by Richyboy170 in PR83; that acceptance does not turn earlier local output into a fresh main run.

Actual implemented paths

Server files below: server/tests/lab-04/; client: client/tests/lab-04/; browser: client/e2e/lab-04/.

ID / AC	Assertions	Actual files
ACTION / AC-01, AC-02, AC-03	Validation, actor vs assignee, immutable events, status/edit rules	action-validation.unit.test.ts, actions-taken.api.test.ts, ActionsTaken.test.tsx, actions-api.test.tsx, actions-taken-flow.spec.ts
AUTH / AC-04	Owned read-only Actions, other-owner404, wrong-role writes denied, private Notes excluded	actions-taken.api.test.ts, ticket-workflow.api.test.ts, actions-taken-flow.spec.ts; existing Lab2/3 regression
WORKFLOW / AC-05, AC-06	Full8x8 matrix for Staff/Admin, unfinished-work/follow-up gates, advisory Requester indication, immutable status history	ticket-workflow.unit.test.ts, ticket-workflow.api.test.ts, TicketWorkflow.test.tsx, workflow-api.test.tsx, ticket-resolution.spec.ts
REQUESTER / AC-07	Owned metrics/calendar boundaries/legacy dates/recents/attention/drill-down; loading/zero/error/403	dashboard-policy.unit.test.ts, dashboards.api.test.ts, requester-dashboard.api.test.ts, RequesterDashboard.test.tsx, dashboards.spec.ts
STAFF / AC-08	Status/priority/unassigned/owner queues, performedBy vs assignedTo, Admin boundaries, matching drill-down	dashboards.api.test.ts, staff-dashboard.api.test.ts, StaffDashboard.test.tsx, RequesterDashboard.test.tsx (shared cases), dashboards.spec.ts
MIGRATION / AC-09	Original data/relations survive additive migration; real pg_dump/restore recovery	data-contract.test.ts, migration-regression.api.test.ts
SEED / AC-10	Repeat seed retains IDs/work/events/versions; zero/one/multiple Actions and distinct actors	seed-regression.api.test.ts
CONCURRENCY / AC-11	Parent locks, version gates, one winner409, Action vs Resolve, replay and rollback	versioned-mutations.integration.test.ts, ticket-workflow.api.test.ts, actions-taken.api.test.ts
RECOVERY / AC-12	Persistent scoped Action key, same-payload replay/different409; conflict/input retention; Comment/Note serialized and cleared only after confirmed save	actions-taken.api.test.ts, ActionsTaken.test.tsx, TicketWorkflow.test.tsx, StaffDraftRecovery.test.tsx, responsive.spec.ts
STYLE / AC-13	One current nav destination, skip link/focus, semantic errors/read-only/non-color cues, axe WCAG2A/AA+2.1AA at three widths	ZenGreenStyle.test.tsx, responsive.spec.ts

ID / AC	Assertions	Actual files
PERF / AC-07, AC-08	10k Tickets/20k Actions/events, five warmups+20HTTP samples/role, p95<=1000ms, bounded inner-model calls and EXPLAIN ANALYZE BUFFERS	dashboard-performance.test.ts (explicit opt-in)

Retained actual local outputs

All paths in artifacts/lab-04/phase7/:

- server-final-consistent-fixture-output.txt:29 files/272 tests passed, opted-in performance included; synthetic Action fixtures retain their paired events.

- client-final-output.txt:19 files/90 tests passed after all semantic follow-ups; client-build-output.txt passes. Unsupported RTL exact option caught earlier by TS.

- e2e-final-verified-output.txt:37 unique prior Lab2/3 and Lab4 scenarios passed on final product source. The10-scenario visual/console subset yields48 axe reports, no automatic violations and no unexpected console/runtime errors; incomplete native-select contrast manually reviewed.64 actual images retained and inspected.

- Staff draft RED (lost input/duplicate requests) and navigation RED retained; focused GREEN proves the fixes. Tooling/parser failures are not behavioral TDD.

- Latest client/server audit JSON:0 advisories after compatible patches and explicitly selected supported tooling. Audit0 is not a full security proof.

Reproduction and limits

See README and phase7-qa.md. Backend test:lab4-qa requires explicit loopback55484 regression DB, admin URL and performance opt-in. Browser test:e2e uses safe playwright.lab4-qa.config.ts, LAB4_UI_TEST=1 and lab4_p7_ui_* DB; reserved API/UI ports55485/55486 with reuseExistingServer=false. Helpers reject other environments before mutation. Only synthetic accounts/data; retain append-only histories.

Performance threshold is a local smoke criterion, not a production SLA. Instrumentation counts inner dashboard model operations, not outer auth SQL or BEGIN/COMMIT. Bulk synthetic fixture insertion is not API execution proof. Axe plus selected keyboard checks is not screen-reader certification. All captures require visual inspection before selecting report figures; current inventory and manual inspection notes are in phase7-visual-evidence.md and phase7-qa.md. Clean lockfile reproducibility is documented above. Fresh final-main output and rendered PDF pagination are separate pending gates; no unexecuted or skipped suite is represented as passing.

Phase8 supplementary proof and reproduction

Staff-specific real empty-personal-work, held-request loading, labelled injected 503/403 and actual200 retry captures are under artifacts/lab-04/phase8/screenshots. They supplement, not relabel, earlier Requester state proof. The capture CLI script rejects non-reserved UI origins; a new synthetic Staff account has no personal work. The working Docker database and append-only histories are untouched.

Installed Chrome exposed a missing favicon request not requested by the earlier headless run. A native SVG favicon/reference fixes that concrete404. Actual GET on the reserved UI origin returns200/image-svg; the Phase8 client/build/full-browser rerun is retained separately under artifacts/lab-04/phase8, never called main/CI. The complete compact backend rerun uses the same environment guard and full test set with the default reporter; verbosity is presentation, not test exclusion.

Complete Phase8 local server output (unit, API, authorization, migration, concurrency, performance).

Exact source / complete raw output

```
RUN  v4.1.12 D:/Chat/cpe/lab4/server
```

```
stdout | tests/lab-04/seed-regression.api.test.ts > SEED-01: migration deployment, constraints and repeated
seeds > deploys again without pending migrations and exposes expected indexes
SEED-01 PASS: all eight migrations deploy; repeated seed preserves work/history; statuses/priorities and
zero/one/multiple Actions covered.

 PASS tests/lab-04/seed-regression.api.test.ts (6 tests) 18921ms
     PASS reruns without duplicates, resetting user changes, passwords or Action history  11245ms
     PASS deploys again without pending migrations and exposes expected indexes  802ms
stdout | tests/lab-04/ticket-workflow.api.test.ts > Phase5 real Ticket workflow HTTP > rejects all three
Resolve gate reasons and Cancel with open work without partial changes
WORKFLOW_GATE_PROOF {"reason":"NO_COMPLETED_ACTION","status":409,"body":{"error":"Unable to complete this
Action request.","code":"RESOLUTION_GATE","fields":{"reason":"NO_COMPLETED_ACTION"}}}

stdout | tests/lab-04/ticket-workflow.api.test.ts > Phase5 real Ticket workflow HTTP > rejects all three
Resolve gate reasons and Cancel with open work without partial changes
WORKFLOW_GATE_PROOF {"reason":"NO_COMPLETED_ACTION","status":409,"body":{"error":"Unable to complete this
Action request.","code":"RESOLUTION_GATE","fields":{"reason":"NO_COMPLETED_ACTION"}}}

stdout | tests/lab-04/ticket-workflow.api.test.ts > Phase5 real Ticket workflow HTTP > rejects all three
Resolve gate reasons and Cancel with open work without partial changes
WORKFLOW_GATE_PROOF {"reason":"FOLLOW_UP_PENDING","status":409,"body":{"error":"Unable to complete this Action
 request.","code":"RESOLUTION_GATE","fields":{"reason":"FOLLOW_UP_PENDING"}}}

stdout | tests/lab-04/ticket-workflow.api.test.ts > Phase5 real Ticket workflow HTTP > rejects all three
Resolve gate reasons and Cancel with open work without partial changes
WORKFLOW_GATE_PROOF {"reason":"OPEN_ACTIONS","resolve":{"status":409,"body":{"error":"Unable to complete this
Action request.","code":"RESOLUTION_GATE","fields":{"reason":"OPEN_ACTIONS"}}},"cancel":{"status":409,"body":{
"error":"Unable to complete this Action request.","code":"OPEN_ACTIONS"}}}

stdout | tests/lab-04/ticket-workflow.api.test.ts > Phase5 real Ticket workflow HTTP > serializes actual HTTP
Resolve against create without bypass
WORKFLOW_RACE_PROOF {"kind":"create","round":0,"statuses":[200,409],"final":"RESOLVED","version":3}

stdout | tests/lab-04/ticket-workflow.api.test.ts > Phase5 real Ticket workflow HTTP > serializes actual HTTP
Resolve against create without bypass
WORKFLOW_RACE_PROOF {"kind":"create","round":1,"statuses":[200,409],"final":"RESOLVED","version":3}

stdout | tests/lab-04/ticket-workflow.api.test.ts > Phase5 real Ticket workflow HTTP > serializes actual HTTP
Resolve against create without bypass
WORKFLOW_RACE_PROOF {"kind":"create","round":2,"statuses":[200,409],"final":"RESOLVED","version":3}

stdout | tests/lab-04/ticket-workflow.api.test.ts > Phase5 real Ticket workflow HTTP > serializes actual HTTP
Resolve against follow-up without bypass
WORKFLOW_RACE_PROOF {"kind":"follow-up","round":0,"statuses":[200,409],"final":"RESOLVED","version":3}

stdout | tests/lab-04/ticket-workflow.api.test.ts > Phase5 real Ticket workflow HTTP > serializes actual HTTP
Resolve against follow-up without bypass
WORKFLOW_RACE_PROOF {"kind":"follow-up","round":1,"statuses":[200,409],"final":"RESOLVED","version":3}

stdout | tests/lab-04/ticket-workflow.api.test.ts > Phase5 real Ticket workflow HTTP > serializes actual HTTP
Resolve against follow-up without bypass
WORKFLOW_RACE_PROOF {"kind":"follow-up","round":2,"statuses":[200,409],"final":"RESOLVED","version":3}

stdout | tests/lab-04/ticket-workflow.api.test.ts > Phase5 real Ticket workflow HTTP > serializes actual HTTP
Resolve against transition without bypass
WORKFLOW_RACE_PROOF {"kind":"transition","round":0,"statuses":[200,409],"final":"RESOLVED","version":3}

stdout | tests/lab-04/ticket-workflow.api.test.ts > Phase5 real Ticket workflow HTTP > serializes actual HTTP
Resolve against transition without bypass
WORKFLOW_RACE_PROOF {"kind":"transition","round":1,"statuses":[200,409],"final":"RESOLVED","version":3}

stdout | tests/lab-04/ticket-workflow.api.test.ts > Phase5 real Ticket workflow HTTP > serializes actual HTTP
Resolve against transition without bypass
WORKFLOW_RACE_PROOF {"kind":"transition","round":2,"statuses":[200,409],"final":"RESOLVED","version":3}

 PASS tests/lab-04/ticket-workflow.api.test.ts (73 tests) 15509ms
     PASS rejects all three Resolve gate reasons and Cancel with open work without partial changes  369ms
     PASS keeps Requester indication advisory and scopes immutable status history to owned Tickets  351ms
     PASS serializes actual HTTP Resolve against create without bypass  383ms
stdout | tests/lab-04/actions-taken.api.test.ts > Lab4 real Action API / authentication / history / replay >
allows only owned Requester reads and never shares Notes, credentials or other-parent history
NEGATIVE_API_PROOF {"case":"cross-requester","pathSuffix":"/actions","status":404,"body":{"error":"Unable to
complete this Action request.","code":"NOT_FOUND"}}

stdout | tests/lab-04/actions-taken.api.test.ts > Lab4 real Action API / authentication / history / replay >
allows only owned Requester reads and never shares Notes, credentials or other-parent history
NEGATIVE_API_PROOF {"case":"cross-
requester","pathSuffix":"/actions/14/events","status":404,"body":{"error":"Unable to complete this Action
request.","code":"NOT_FOUND"}}
```

```
stdout | tests/lab-04/actions-taken.api.test.ts > Lab4 real Action API / authentication / history / replay >
allows only owned Requester reads and never shares Notes, credentials or other-parent history
NEGATIVE_API_PROOF {"case":"cross-requester","pathSuffix":"/workflow-
events","status":404,"body":{"error":"Unable to complete this Action request.","code":"NOT_FOUND"}}

stdout | tests/lab-04/actions-taken.api.test.ts > Lab4 real Action API / authentication / history / replay >
replays a persistent normalized payload before stale checks with no duplicate work/history
NEGATIVE_API_PROOF {"case":"idempotency-payload-mismatch","status":409,"body":{"error":"Unable to complete
this Action request.","code":"IDEMPOTENCY_CONFLICT"}}

stdout | tests/lab-04/actions-taken.api.test.ts > Lab4 real Action API / authentication / history / replay >
prevents the legacy status writer bypassing Action gates or shared versions
NEGATIVE_API_PROOF {"case":"legacy-resolution-bypass","status":409,"body":{"error":"Unable to complete this
Action request.","code":"RESOLUTION_GATE","fields":{"reason":"NO_COMPLETED_ACTION"}}}

stdout | tests/lab-04/actions-taken.api.test.ts > Lab4 real Action API / authentication / history / replay >
serializes actual HTTP Resolve vs Action creation, then retains immutable workflow history
CONCURRENCY_API_PROOF {"case":"Resolve-vs-Action","statuses":[200,409],"finalStatus":"RESOLVED","version":6}

 PASS tests/lab-04/actions-taken.api.test.ts (14 tests) 11049ms
     PASS edits, assigns and transitions atomically, retaining before/after audit and terminal rules  506ms
     PASS enforces merged follow-up/result values and stale edit checks without bumping versions  349ms
     PASS returns stable paginated lists, safe own-created work, and rejects invalid paging  803ms
     PASS serializes actual HTTP Resolve vs Action creation, then retains immutable workflow history  410ms
stdout | tests/lab-04/dashboard-performance.test.ts > PERF-01 isolated 10k Tickets /20k Actions smoke >
measures real authorized dashboards, fixed query work and query plans
PERF_ENV {"node":"v22.17.0","platform":"win32","arch":"x64","cpu":"Intel(R) Core(TM)
i7-14650HX","logicalCpus":24,"totalMemoryGB":24,"postgres":"dedicated loopback55484","database":"/lab4_p2_perf
_4a4ed978bc404017a79f208f1b1309e3","warmup":5,"samples":20,"tickets":10000,"actions":20000,"limitMs":1000}

stdout | tests/lab-04/dashboard-performance.test.ts > PERF-01 isolated 10k Tickets /20k Actions smoke >
measures real authorized dashboards, fixed query work and query plans
PERF_RESULT {"role":"REQUESTER","path":"/api/tickets/dashboard","timesMs":[44.62,42.62,43.03,60.91,52.07,53.81
,42.73,45.15,43.81,48.78,45.97,43.26,44.28,45.49,48.73,43.26,46.57,45.07,37.58,52.12],"p95Ms":53.81,"maxMs":60
.91,"transactionModelOperations":[6,6,6,6,6,6,6,6,6,6,6,6,6,6,6,6,6,6,6,6],"responseBytes":2362}

stdout | tests/lab-04/dashboard-performance.test.ts > PERF-01 isolated 10k Tickets /20k Actions smoke >
measures real authorized dashboards, fixed query work and query plans
PERF_RESULT {"role":"IT_STAFF","path":"/staff/dashboard","timesMs":[113.27,116.03,112.69,116.06,113.42,115.49,
115.09,113,113.17,113.75,109.38,119.59,114.56,114.81,113.74,114.92,113.2,118.79,107.2,110.39],"p95Ms":118.79,"
maxMs":119.59,"transactionModelOperations":[10,10,10,10,10,10,10,10,10,10,10,10,10,10,10,10,10,10,10,10],"resp
onseBytes":3894,"myActions":20000}

stdout | tests/lab-04/dashboard-performance.test.ts > PERF-01 isolated 10k Tickets /20k Actions smoke >
measures real authorized dashboards, fixed query work and query plans
PERF_RESULT {"role":"ADMINISTRATOR","path":"/staff/dashboard","timesMs":[80.08,79.43,91.85,80.14,78.96,93.28,7
5.11,79.95,82.93,80,80.3,77.87,79.13,89.36,79.86,83.58,85.83,80.8,81.43,75.57],"p95Ms":91.85,"maxMs":93.28,"tr
ansactionModelOperations":[10,10,10,10,10,10,10,10,10,10,10,10,10,10,10,10,10,10,10,10],"responseBytes":3096,"
myActions":0}
PERF_EXPLAIN SELECT "currentStatus",count(*) FROM "Ticket" GROUP BY "currentStatus"
HashAggregate  (cost=350.00..350.08 rows=8 width=12) (actual time=5.304..5.308 rows=8.00 loops=1)
  Group Key: "currentStatus"
  Batches: 1  Memory Usage: 32kB
  Buffers: shared hit=200
  ->  Seq Scan on "Ticket"  (cost=0.00..300.00 rows=10000 width=4) (actual time=0.039..3.236 rows=10000.00
loops=1)
        Buffers: shared hit=200
Planning:
  Buffers: shared hit=215
Planning Time: 8.140 ms
Execution Time: 5.424 ms
PERF_EXPLAIN SELECT id FROM "Ticket" WHERE "requesterUserId"=1 ORDER BY "updatedAt" DESC,id DESC LIMIT 5
Limit  (cost=491.10..491.11 rows=5 width=12) (actual time=3.558..3.559 rows=5.00 loops=1)
  Buffers: shared hit=206
  ->  Sort  (cost=491.10..516.10 rows=10000 width=12) (actual time=3.556..3.556 rows=5.00 loops=1)
        Sort Key: "updatedAt" DESC, id DESC
        Sort Method: top-N heapsort  Memory: 25kB
        Buffers: shared hit=206
        ->  Seq Scan on "Ticket"  (cost=0.00..325.00 rows=10000 width=12) (actual time=0.027..1.997
rows=10000.00 loops=1)
              Filter: ("requesterUserId" = 1)
              Buffers: shared hit=200
Planning:
  Buffers: shared hit=213
Planning Time: 4.827 ms
Execution Time: 3.589 ms
PERF_EXPLAIN SELECT id FROM "ActionTaken" WHERE "performedByUserId"=2 ORDER BY "createdAt" DESC,id DESC LIMIT
5
```

```
Limit  (cost=868.19..868.21 rows=5 width=12) (actual time=6.677..6.679 rows=5.00 loops=1)
  Buffers: shared hit=292
  ->  Sort  (cost=868.19..918.19 rows=20000 width=12) (actual time=6.675..6.676 rows=5.00 loops=1)
        Sort Key: "createdAt" DESC, id DESC
        Sort Method: top-N heapsort  Memory: 25kB
        Buffers: shared hit=292
        ->  Seq Scan on "ActionTaken"  (cost=0.00..536.00 rows=20000 width=12) (actual time=0.017..3.589
rows=20000.00 loops=1)
              Filter: ("performedByUserId" = 2)
              Buffers: shared hit=286
Planning:
  Buffers: shared hit=156
Planning Time: 3.187 ms
Execution Time: 6.704 ms

 PASS tests/lab-04/dashboard-performance.test.ts (1 test) 9079ms
     PASS measures real authorized dashboards, fixed query work and query plans  9072ms
 PASS tests/lab-04/versioned-mutations.integration.test.ts (14 tests) 7630ms
stdout | tests/lab-04/migration-regression.api.test.ts > MIG-01: populated Lab3 upgrade and recovery on
isolated PostgreSQL > restores the pre-upgrade backup into another fresh database with exact equality
MIG-01 PASS: exact nine-table equality, legacy zero-Action/NULL resolution, separate-database
pg_dump/pg_restore recovery.

 PASS tests/lab-04/migration-regression.api.test.ts (3 tests) 4994ms
     PASS preserves every existing field/ID/relationship in all nine tables  871ms
     PASS restores the pre-upgrade backup into another fresh database with exact equality  1689ms
 PASS tests/lab-03/auth.api.test.ts (3 tests) 3003ms
     PASS creates a session for a valid active User and revokes it on logout  441ms
     PASS uses the same safe response for invalid credentials and inactive accounts  319ms
     PASS requires the current password, clears first-login state, and revokes the old session  2238ms
stdout | tests/lab-04/dashboards.api.test.ts > Phase6 real dashboard metrics, identity and drill-down >
returns numeric zeros, every enum bucket and empty lists before fixtures exist
DASHBOARD_ZERO_PROOF {"requester":{"openTickets":0,"waitingTickets":0,"recentlyResolvedTickets":0},"staff":{"u
nassignedOpenTickets":0,"myOpenTickets":0,"ticketsByStatus":{"NEW":0,"OPEN":0,"IN_PROGRESS":0,"WAITING_FOR_REQ
UESTER":0,"RESOLVED":0,"CLOSED":0,"REOPENED":0,"CANCELLED":0},"openTicketsByPriority":{"NOT_SET":0,"LOW":0,"ME
DIUM":0,"HIGH":0,"URGENT":0},"myActions":0}}

stdout | tests/lab-04/dashboards.api.test.ts > Phase6 real dashboard metrics, identity and drill-down >
computes exact populations and matches every published drill-down against independent fixture predicates
DASHBOARD_QUERY_PROOF {"asOf":"2026-10-29T17:00:00.000Z","window":{"start":"2026-10-
23T17:00:00.000Z","end":"2026-10-
29T17:00:00.000Z"},"requester":{"openTickets":7,"waitingTickets":1,"recentlyResolvedTickets":2},"staff":{"unas
signedOpenTickets":5,"myOpenTickets":2,"ticketsByStatus":{"NEW":1,"OPEN":2,"IN_PROGRESS":1,"WAITING_FOR_REQUES
TER":2,"RESOLVED":3,"CLOSED":2,"REOPENED":2,"CANCELLED":1},"openTicketsByPriority":{"NOT_SET":2,"LOW":1,"MEDIU
M":1,"HIGH":1,"URGENT":3},"myActions":1},"allDrillDownCountsMatch":true}

 PASS tests/lab-04/dashboards.api.test.ts (7 tests) 1860ms
 PASS tests/lab-04/requester-dashboard.api.test.ts (2 tests) 1492ms
 PASS tests/lab-03/users-admin.api.test.ts (5 tests) 1493ms
     PASS creates, searches, edits, and resets a managed user  664ms
 PASS tests/lab-03/requester-authorization.api.test.ts (4 tests) 1358ms
     PASS requires a session and rejects staff or first-login Requesters  373ms
 PASS tests/lab-04/staff-dashboard.api.test.ts (2 tests) 1425ms
 PASS tests/lab-03/staff-ticket-detail.api.test.ts (4 tests) 1379ms
 PASS tests/lab-03/staff-queue.api.test.ts (3 tests) 1240ms
 PASS tests/lab-02/attachments.api.test.ts (5 tests) 998ms
 PASS tests/lab-02/ticket-detail.api.test.ts (3 tests) 899ms
 PASS tests/lab-03/migration-regression.api.test.ts (3 tests) 747ms
     PASS dual-writes a legacy requester Ticket to its migrated User during the transition  554ms
 PASS tests/lab-02/create-ticket.api.test.ts (2 tests) 534ms
 PASS tests/lab-02/my-tickets.api.test.ts (2 tests) 470ms
 PASS tests/lab-03/requester-session.test.ts (1 test) 455ms
     PASS revokes only the sessions created by the test helper  454ms
 PASS tests/lab-03/seed-ticket-examples.api.test.ts (2 tests) 290ms
 PASS tests/lab-01/categories.test.ts (1 test) 73ms
 PASS tests/lab-02/development-requesters.api.test.ts (1 test) 22ms
 PASS tests/lab-01/health.test.ts (1 test) 30ms
 PASS tests/lab-04/dashboard-policy.unit.test.ts (12 tests) 11ms
 PASS tests/lab-04/ticket-workflow.unit.test.ts (65 tests) 9ms
 PASS tests/lab-04/action-validation.unit.test.ts (30 tests) 13ms
 PASS tests/lab-04/data-contract.test.ts (1 test) 5ms
 PASS tests/lab-02/ticket-number.test.ts (2 tests) 5ms

 Test Files  29 passed (29)
      Tests  272 passed (272)
   Start at  22:52:32
   Duration  99.04s (transform 1.44s, setup 0ms, import 8.31s, tests 84.99s, environment 3ms)
```

Complete Phase8 local client run; not final main.

Exact source / complete raw output

```
> toktickit-client@1.0.0 test
> vitest run --reporter=default


 RUN  v4.1.12 D:/Chat/cpe/lab4/client

 PASS tests/lab-04/workflow-api.test.tsx (7 tests) 252ms
 PASS tests/lab-04/actions-api.test.tsx (9 tests) 257ms
 PASS tests/lab-02/ZenGreenStyle.test.tsx (1 test) 1115ms
     PASS uses labelled required fields and a disabled-safe submit state  1085ms
 PASS tests/lab-03/RoleBadge.test.tsx (4 tests) 1284ms
     PASS renders REQUESTER with a text-bearing badge  472ms
     PASS uses role badges in both the desktop user table and mobile cards  630ms
 PASS tests/lab-04/ZenGreenStyle.test.tsx (1 test) 2149ms
   PASS exposes exactly one current-page navigation action and follows user navigation  2095ms
 PASS tests/lab-04/StaffDraftRecovery.test.tsx (3 tests) 3072ms
     PASS retains Add Public Comment through failure then clears only after confirmed success  1655ms
     PASS retains Add Internal Note through failure then clears only after confirmed success  772ms
     PASS prevents repeated posts and protects input while the original request is pending  616ms
 PASS tests/lab-01/App.test.tsx (4 tests) 3124ms
     PASS renders the TokTickIT heading  608ms
     PASS shows a loading state while the API request is pending  1393ms
     PASS shows Online and the categories returned by the API  733ms
     PASS shows an Offline error message when the API is unavailable  356ms
 PASS tests/lab-04/RequesterDashboard.test.tsx (14 tests) 3396ms
     PASS shows loading, then numeric zeros and explicit empty lists  1540ms
     PASS passes the exact owned status-group predicate to the list  717ms
     PASS shows safe failure rather than fabricated zero totals; retries successfully  446ms
     PASS refreshes the snapshot after a new server response  606ms
 PASS tests/lab-03/StaffTicketQueue.test.tsx (3 tests) 3608ms
     PASS renders distinct requested and IT priority badges in table and mobile cards  1158ms
     PASS renders queue controls and an explicit no-results state  477ms
     PASS sends a search query and distinguishes filtered no-results  1933ms
 PASS tests/lab-04/StaffDashboard.test.tsx (4 tests) 4001ms
   PASS shows correct IT_STAFF heading and zero enum buckets  2553ms
   PASS shows correct ADMINISTRATOR heading and zero enum buckets  796ms
   PASS carries session-owner drill-down and current-performer My Actions as distinct populations  541ms
 PASS tests/lab-02/MyTickets.test.tsx (2 tests) 4045ms
     PASS shows owned Ticket data and sends search/filter query controls  3078ms
     PASS distinguishes an empty Ticket list from no matching results  941ms
 PASS tests/lab-02/DevelopmentRequester.test.tsx (4 tests) 4326ms
     PASS requires login and never offers a Development Requester selector  1391ms
     PASS uses the authenticated User for the requester shell  2550ms
 PASS tests/lab-03/StaffTicketDetail.test.tsx (3 tests) 4703ms
     PASS separates Public Comments from Internal Notes and exposes staff controls  2195ms
     PASS posts a Public Comment through the staff API  1907ms
     PASS allows Administrator IT Priority changes while keeping staff-only controls unavailable  563ms
 PASS tests/lab-03/Authentication.test.tsx (2 tests) 4952ms
     PASS submits credentials and returns the authenticated user  3332ms
     PASS keeps a mismatched password in the form and explains the problem  1590ms
 PASS tests/lab-02/CreateTicket.test.tsx (2 tests) 6061ms
     PASS shows field-level validation without calling the API  1343ms
     PASS submits valid input and displays the backend Ticket Number  4703ms
 PASS tests/lab-04/TicketWorkflow.test.tsx (9 tests) 5846ms
     PASS makes Requester history read-only and uses owned routes only  476ms
     PASS blocks Resolve with explicit missing work/follow-up reasons without sending a write  1452ms
     PASS offers only matrix destinations and requires explicit keyboard-focusable confirmation for
Administrator  1194ms
     PASS preserves a failed selection and version through recoverable503  766ms
     PASS requires an actual refresh and deliberate version adoption after409  968ms
     PASS retains selection but disables writes when refreshed saved status makes it illegal  316ms
     PASS prevents double submits while status update is pending  374ms
 PASS tests/lab-02/RequesterTicketDetail.test.tsx (4 tests) 6125ms
     PASS renders owner-scoped Ticket fields read-only and returns to My Tickets  3150ms
     PASS posts a Public Comment and records a non-final resolution indication  1596ms
     PASS uploads an allowed Attachment and requires a reason before soft removal  1313ms
 PASS tests/lab-03/UserManagement.test.tsx (4 tests) 7042ms
     PASS replaces edit fields when switching directly to another account  3967ms
     PASS lists users with filters and management actions  355ms
     PASS edits a user's role and active state  999ms
     PASS creates a user and resets an initial password  1704ms
 PASS tests/lab-04/ActionsTaken.test.tsx (10 tests) 9244ms
     PASS renders separate performer/assignee and read-only Requester history without fetching staff data
2664ms
```

```
     PASS validates required description and conditional follow-up note near fields without an API call
2369ms
     PASS retains the draft and same logical retry key through a recoverable API failure  1088ms
     PASS requires deliberate refresh/reconcile for a stale version while preserving edited values  1101ms
     PASS shows inactive-assignee server validation without clearing the form  628ms
     PASS freezes status-confirmation versions even if the saved Ticket is refreshed before confirmation
370ms
     PASS prevents duplicate submissions while the original create request is pending  636ms

 Test Files  19 passed (19)
      Tests  90 passed (90)
   Start at  22:44:41
   Duration  23.61s (transform 27.93s, setup 48.21s, import 98.69s, tests 74.60s, environment 96.97s)
```

Complete Phase8 local browser regression; not final main.

Exact source / complete raw output

```
> toktickit-client@1.0.0 test:e2e
> playwright test --config playwright.lab4-qa.config.ts

[WebServer] (node:60856) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
[WebServer] (Use `node --trace-warnings ...` to show where the warning was created)
[WebServer] (node:30344) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
[WebServer] (Use `node --trace-warnings ...` to show where the warning was created)
[WebServer] (node:13480) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
[WebServer] (Use `node --trace-warnings ...` to show where the warning was created)
[WebServer] (node:13480) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
[WebServer] (Use `node --trace-warnings ...` to show where the warning was created)
[WebServer] (node:11368) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
[WebServer] (Use `node --trace-warnings ...` to show where the warning was created)
[WebServer] (node:66776) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
[WebServer] (Use `node --trace-warnings ...` to show where the warning was created)

Running 37 tests using 1 worker

(node:57964) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
(Use `node --trace-warnings ...` to show where the warning was created)
  ok  1 e2e\lab-02\requester-ticket-flow.spec.ts:21:1 › Requester creates, finds, and opens an owned Ticket
(2.7s)
  ok  2 e2e\lab-02\requester-ticket-flow.spec.ts:38:1 › cross-owner access is blocked while the owner can
upload and remove an Attachment (6.0s)
  ok  3 e2e\lab-02\responsive.spec.ts:16:1 › My Tickets remains usable without horizontal overflow at desktop,
 tablet, and mobile widths (3.3s)
  ok  4 e2e\lab-03\authentication.spec.ts:6:1 › first login requires password change and logout blocks
protected access (7.0s)
  ok  5 e2e\lab-03\authentication.spec.ts:7:1 › invalid and inactive credentials use the same safe failure
(2.3s)
  ok  6 e2e\lab-03\responsive.spec.ts:13:1 › major Lab 3 screens have no horizontal document overflow (25.8s)
  ok  7 e2e\lab-03\staff-ticket-flow.spec.ts:8:1 › IT Staff operational workflow preserves shared Comments
versus private Notes (6.2s)
  ok  8 e2e\lab-03\user-administration.spec.ts:7:1 › Administrator changes IT Priority without receiving
staff-only controls (3.0s)
  ok  9 e2e\lab-03\user-administration.spec.ts:31:1 › Administrator manages a user and forces password change
(8.3s)
  ok 10 e2e\lab-04\actions-taken-flow.spec.ts:52:1 › real multi-staff lifecycle: create, edit, assign,
complete/cancel and owned Requester read-only history (12.5s)
  ok 11 e2e\lab-04\actions-taken-flow.spec.ts:99:1 › uncertain transport response retains draft/key; real
backend replay creates only one record (4.2s)
  ok 12 e2e\lab-04\actions-taken-flow.spec.ts:119:1 › real inactive-assignee race retains draft and accepts a
corrected active selection (4.2s)
  ok 13 e2e\lab-04\actions-taken-flow.spec.ts:133:1 › real competing edit returns 409; refresh and explicit
rebase retain local input (4.4s)
  ok 14 e2e\lab-04\actions-taken-flow.spec.ts:151:1 › Administrator creation is allowed and follows the same
Action policy (3.4s)
  ok 15 e2e\lab-04\actions-taken-flow.spec.ts:157:1 › fully loaded forms, validation, keyboard focus and
responsive layout at three widths (4.3s)
  ok 16 e2e\lab-04\actions-taken-flow.spec.ts:174:1 › labelled delayed assignee loading and safe
forbidden/load retry state (3.7s)
  ok 17 e2e\lab-04\dashboards.spec.ts:8:1 › Requester counts and exact open drill-down; detail returns to
refreshed Dashboard (3.1s)
  ok 18 e2e\lab-04\dashboards.spec.ts:17:1 › Staff metrics, own-performer work and all status/priority drill-
downs retain exact query (8.6s)
  ok 19 e2e\lab-04\dashboards.spec.ts:28:1 › Administrator Dashboard is available without granting Requester
Dashboard access (2.6s)
  ok 20 e2e\lab-04\dashboards.spec.ts:29:1 › loading, injected failure and forbidden never display stale
```

```
metrics; deliberate retry works (3.3s)
  ok 21 e2e\lab-04\dashboards.spec.ts:33:1 › zero owned population is a real API result, not a loading
fallback (3.0s)
  ok 22 e2e\lab-04\dashboards.spec.ts:34:1 › three viewport layouts keep dashboard controls within document
width and keyboard focus visible (6.2s)
  ok 23 e2e\lab-04\responsive.spec.ts:36:3 › Requester screens: loaded semantic/contrast/keyboard inspection
at desktop (11.2s)
  ok 24 e2e\lab-04\responsive.spec.ts:44:3 › Staff work forms, confirmations and My Actions inspection at
desktop (20.2s)
  ok 25 e2e\lab-04\responsive.spec.ts:54:3 › Administrator directory and edit/reset forms inspection at
desktop (7.7s)
  ok 26 e2e\lab-04\responsive.spec.ts:36:3 › Requester screens: loaded semantic/contrast/keyboard inspection
at tablet (10.9s)
  ok 27 e2e\lab-04\responsive.spec.ts:44:3 › Staff work forms, confirmations and My Actions inspection at
tablet (19.2s)
  ok 28 e2e\lab-04\responsive.spec.ts:54:3 › Administrator directory and edit/reset forms inspection at tablet
 (7.1s)
  ok 29 e2e\lab-04\responsive.spec.ts:36:3 › Requester screens: loaded semantic/contrast/keyboard inspection
at mobile (10.1s)
  ok 30 e2e\lab-04\responsive.spec.ts:44:3 › Staff work forms, confirmations and My Actions inspection at
mobile (17.8s)
  ok 31 e2e\lab-04\responsive.spec.ts:54:3 › Administrator directory and edit/reset forms inspection at mobile
 (6.4s)
  ok 32 e2e\lab-04\responsive.spec.ts:60:1 › real Staff draft survives a labelled injected failure and is
posted once after retry (3.2s)
  ok 33 e2e\lab-04\ticket-resolution.spec.ts:14:1 › real work gate then Resolve, Close, Administrator Reopen
and Cancel with ordered history (10.2s)
  ok 34 e2e\lab-04\ticket-resolution.spec.ts:27:1 › Requester indication remains advisory; own history read-
only and formal API403 (3.7s)
  ok 35 e2e\lab-04\ticket-resolution.spec.ts:33:1 › real competing Ticket write409 preserves selection until
refresh and explicit rebase (4.2s)
  ok 36 e2e\lab-04\ticket-resolution.spec.ts:39:1 › injected recoverable failure retains selection/version and
 succeeds on deliberate retry (3.3s)
  ok 37 e2e\lab-04\ticket-resolution.spec.ts:45:1 › workflow confirmation and gate remain readable with
keyboard focus at all three widths (3.4s)

  37 passed (4.9m)
```

Phase8 local client production build.

Exact source / complete raw output

```
> toktickit-client@1.0.0 build
> tsc && vite build

vite v6.4.5 building for production...
transforming...
PASS 44 modules transformed.
rendering chunks...
computing gzip size...
dist/index.html                   0.47 kB | gzip:  0.30 kB
dist/assets/index-Bhral_cW.css  231.92 kB | gzip: 31.06 kB
dist/assets/index-CVHkxain.js   246.94 kB | gzip: 70.02 kB
PASS built in 1.08s
```

Phase8 local server production build.

Exact source / complete raw output

```
> toktickit-server@1.0.0 build
> tsc
```

Test counts include preserved earlier labs; they are not all new Lab4 tests. Repeated focused runs are not added to the unique-case total. Earlier failures remain in the evidence bundle. No configured GitHub checks means no CI result, not a CI Pass. Raw source logs are linked alongside every rendered output.

Answer Part 4 - AI Use with Reflection

The student supplied GPT-6.1 Sol (medium) and explicitly confirmed the exact Lab4 reflection on30October2026. Ten authentic selected prompts are rendered without inventing original English dialogue. AI implementation/test assistance is disclosed rather than portrayed as student-only work.

Rendered docs/lab-04/ai-use.md

Exact source / complete raw output

Lab 4 AI Use with Reflection

Tool: OpenAI Codex. Student-reported model: GPT-6.1 Sol (medium), supplied 30October2026. No model name or reflection is copied from Lab3. Earlier contemporaneous entries remain in ai-use-history.md.

Specification agent and coding agent

Specification assistance translated the stakeholder request into explicit FR/BR, status/gate/metric rules, role boundaries, migration decisions, ACs and planned tests before product implementation. The student approved contract decisions. Coding assistance implemented and tested that approved contract, investigated real failures, inspected/captured evidence and prepared peer-reviewed changes. Generated code/test claims were checked against executed results; peer approval and merge belong to actual human reviewers, not the AI.

Ten selected authentic prompts

Original wording is retained, including Thai. English descriptions below explain the assistance; they are not fictitious original prompts. Dates are contemporaneous record dates. Commands to proceed do not themselves prove successful completion.

Date	Original selected prompt	Assistance and verification
2026-10-08	มาเริ่ม lab4 กันเลย	Read the handout/workflow, establish the released baseline and draft the engineering contract/evidence plan.
2026-10-08	I agree your accord	Record student approval of D-01..04; refine contract and test plan before code.
2026-10-16	ทำ Lab4 ของเรากันต่อดีกว่า	Investigate jarbbie's route-contract review; correct Requester /api paths, add source checks and reply in the same PR.
2026-10-28	ถ้า Phase2 เสร็จ เปิด PR ได้เลย	Publish verified data/migration work as PR78; verify direct Issue link, request Richy's review, do not self-merge.
2026-10-28	คุณก็แก้ได้เลย	Address both PR78 version/history findings with atomic mutation primitives and14 actual transaction cases; reply and rerun regression.
2026-10-29	ฉันเห็นละก็โอเคดี ทำขั้นต่อไปได้เลย	Implement authorized Action API/session/ownership/durable retry/history; real negative/race tests catch legacy gate bypass and fixture cleanup issues.
2026-10-29	เริ่ม Phase 4 ได้เลย	Integrate Actions Taken UI; test create/edit/assign/terminal/recovery behavior and inspect loaded3width screenshots.
2026-10-29	เริ่ม Phase5 ได้เลย	Implement approved Ticket matrix/gates/immutable history with real all-pair/role/concurrency/browser verification.

Date	Original selected prompt	Assistance and verification
2026-10-30	ไปทำขั้นต่อไปกันต่อเลย	Implement dashboards with independent DB-count comparison, exact drill-down, Bangkok window/legacy dates and responsive/error/zero tests.
2026-10-30	ปิดงานและทำขั้นต่อไป	Verify Phase6 closure then perform full QA; genuine draft/navigation/semantic RED/GREEN, supported dependency patches, performance and all-screen accessibility evidence.

My Reflection

Confirmed by the student on30October2026 (Asia/Bangkok). In response to the exact three-sentence draft below, the student answered: "ตรงกับประสบการณ์ของฉัน" (it matches my experience). This confirmation applies to Lab4, not Lab3.

Using an AI specification agent helped me turn a broad lab request into explicit rules and acceptance criteria. With the coding agent, I learned to check real test output and evidence rather than assume that generated code or a working screen meant the whole task was complete. The work also reinforced that peer review requires replies and a reviewer-performed merge, not just an approval button.

Actual executed results and limitations are in tests.md/phase7-qa.md; injected failure demonstrations, local runs, peer acceptance and final-main proof remain distinct. No invented dialogue or personal experience is attributed.

Answer Part 5 - Working IT Staff Dashboard UI

Backend snapshots calculate operational status, IT Priority, unassigned/open ownership queues, recent/urgent Tickets and current-user performed Actions. Performed-by is separate from assignee. All19 card/list populations were compared with independent real DB fixture predicates in Phase6; matching drill-down is executable API/browser coverage, not a manually typed metric.

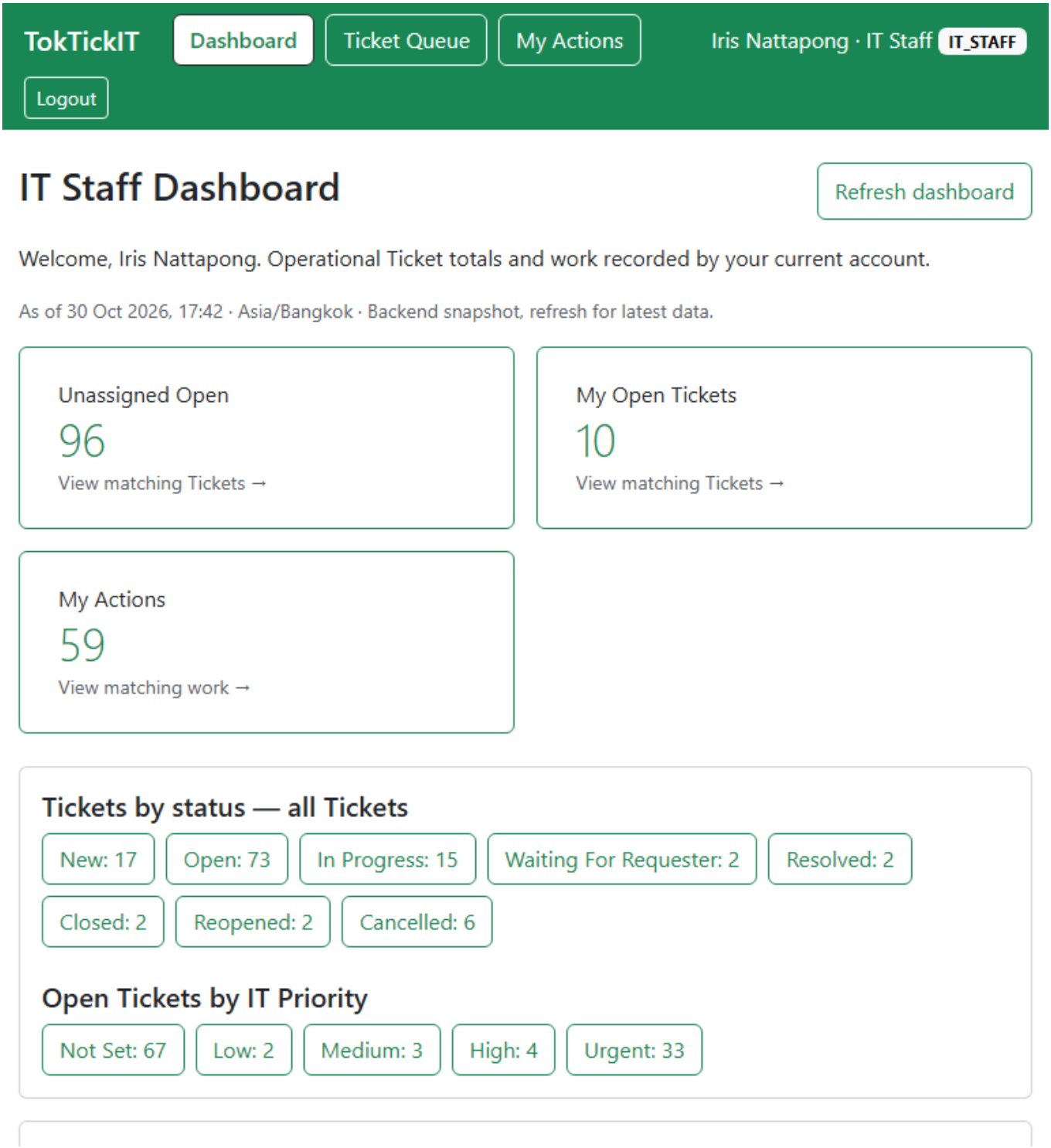


Figure 4. Actual loaded Staff Dashboard at768x1024, with deterministic fixture counts, identity, grouped metrics and destinations. Full long original retained. Top viewport only; full original linked. Full original

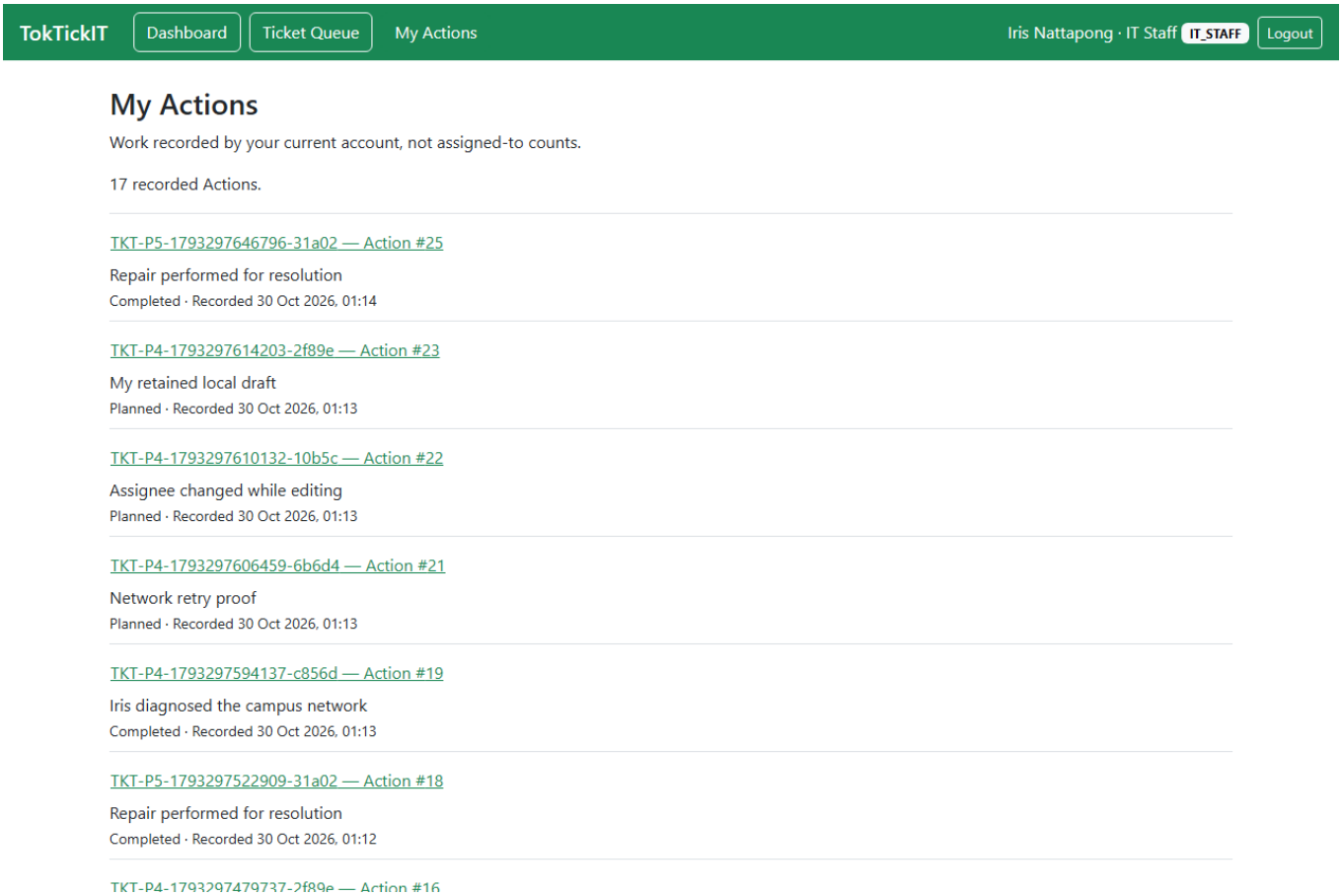


Figure 5. Actual My Actions drill-down uses authenticated performedBy, not assignedTo; rows link back to the corresponding Staff Ticket Detail. Top viewport only; full original linked. Full original

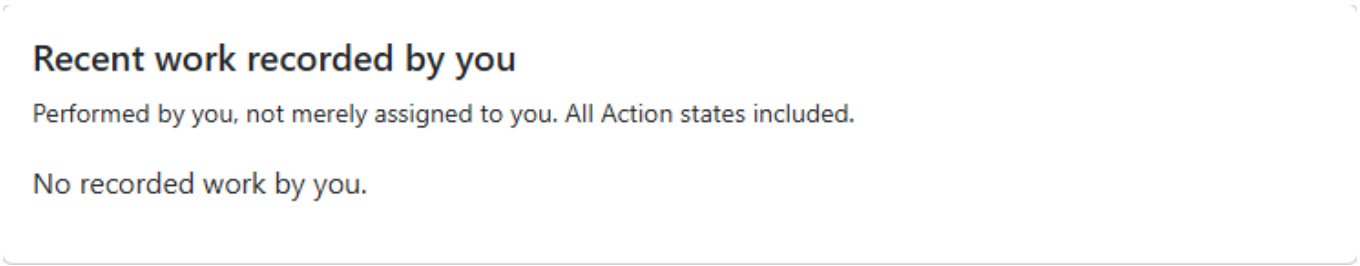


Figure 6. Real new synthetic Staff account with zero personal Actions and owned open Tickets. The global queue is populated; this is NOT a claim that the entire database is empty. Full original

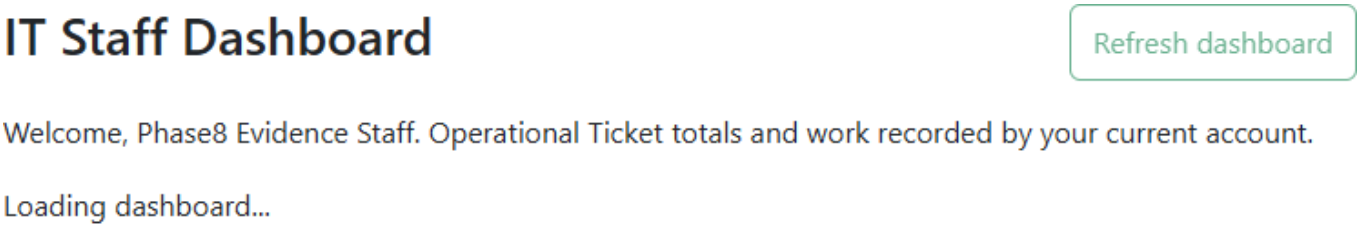


Figure 7. Genuine request held in browser before release: loading state and disabled Refresh. Delay is controlled evidence, not a backend latency measurement. Full original

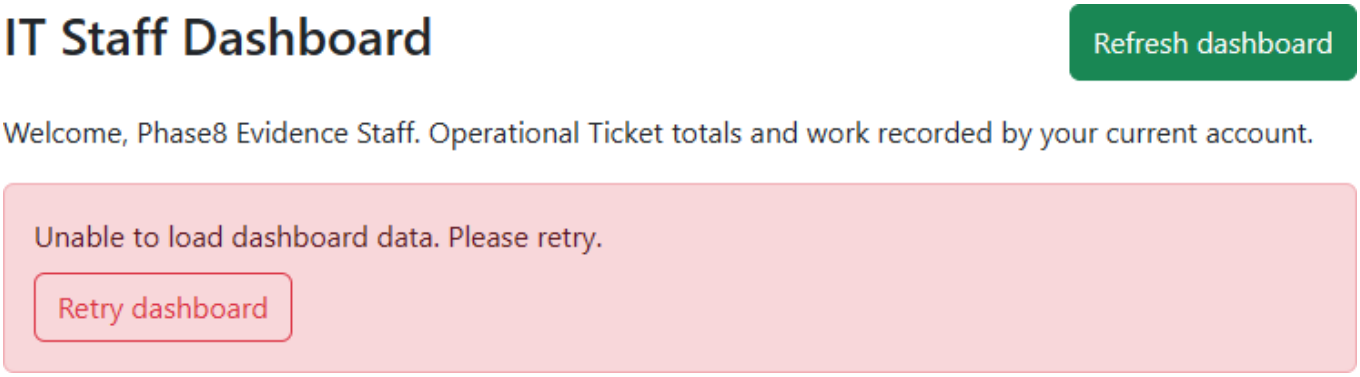


Figure 8. Explicitly injected HTTP503: safe failure and Retry, stale metrics removed. Full original

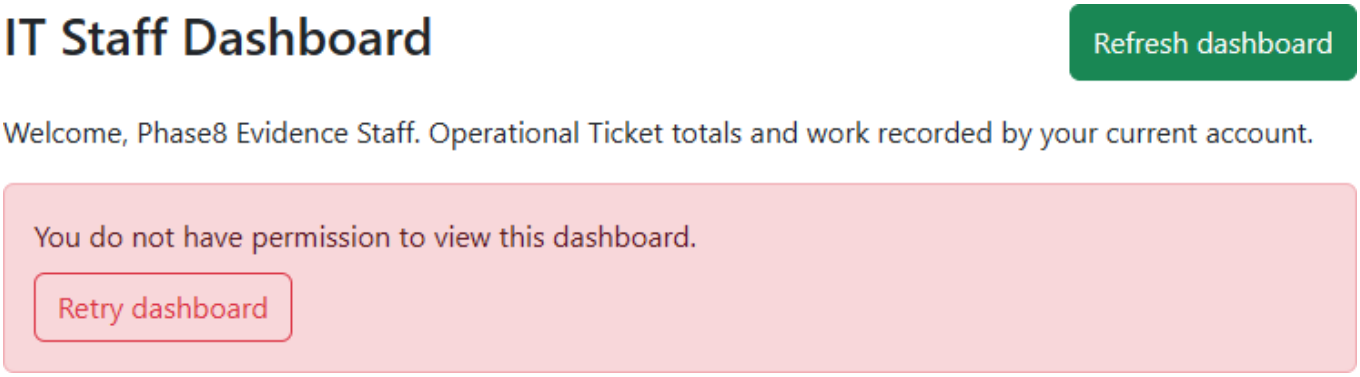


Figure 9. Explicitly injected HTTP403: safe forbidden feedback with no retained metrics. Real role authorization remains separately tested at API level. Full original

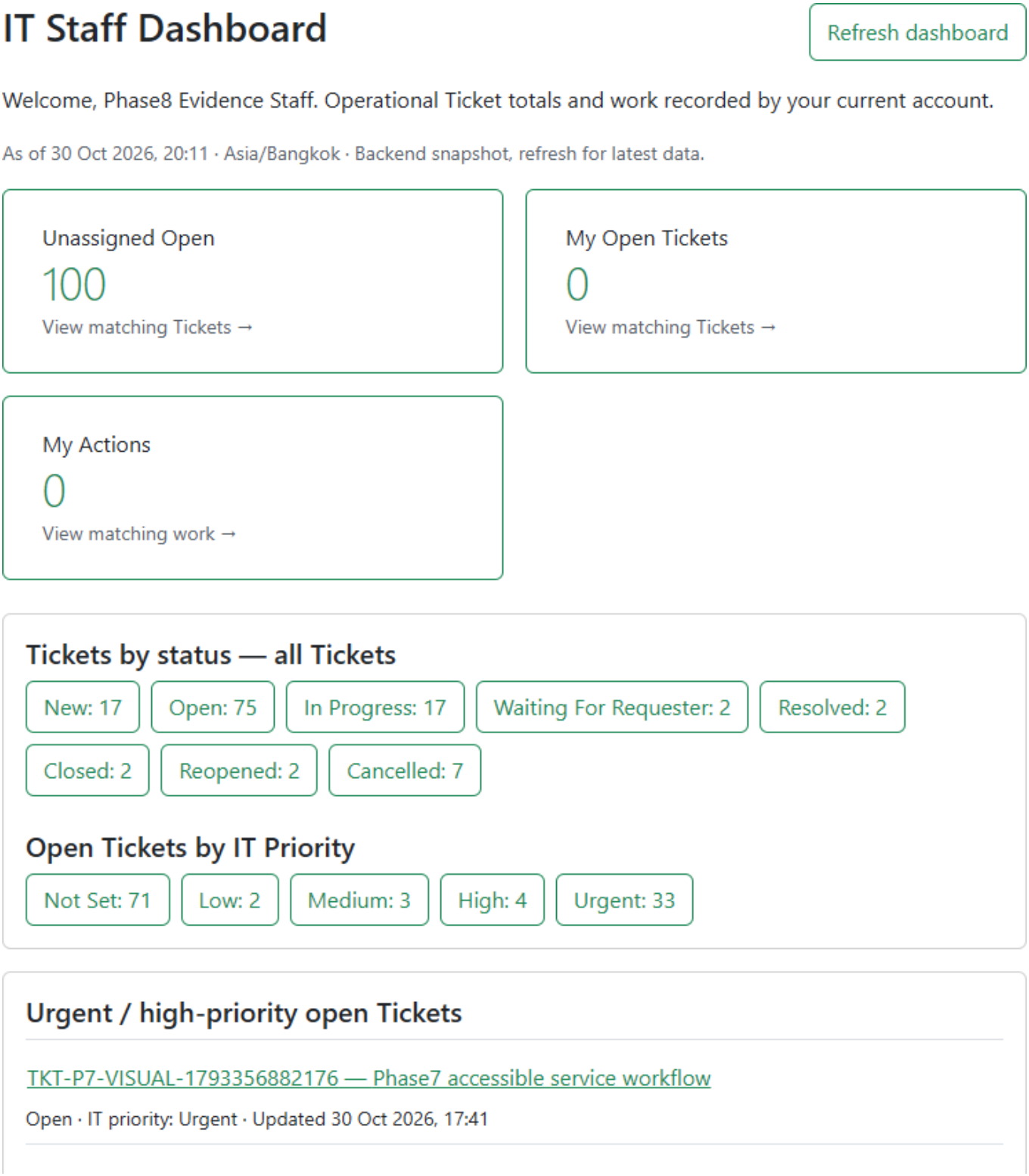


Figure 10. Route injection removed; Retry receives actual HTTP200 and restores the real Dashboard. Top viewport only; full original linked. Full original

Supplemental CLI capture script and structured output: capture script, observations. Synthetic isolated database on127.0.0.1:55484; working Docker database untouched.

Answer Part 6 - Working Actions Taken UI

Actions belong to one Ticket. The authenticated recorded/performed-by actor is immutable and distinct from active Staff/Admin assignee. Create starts Planned; In Progress, Completed and Cancelled transitions follow the approved matrix. Completion requires a Result; follow-up requires a Note. Updates append immutable history and increment Ticket/Action versions atomically.

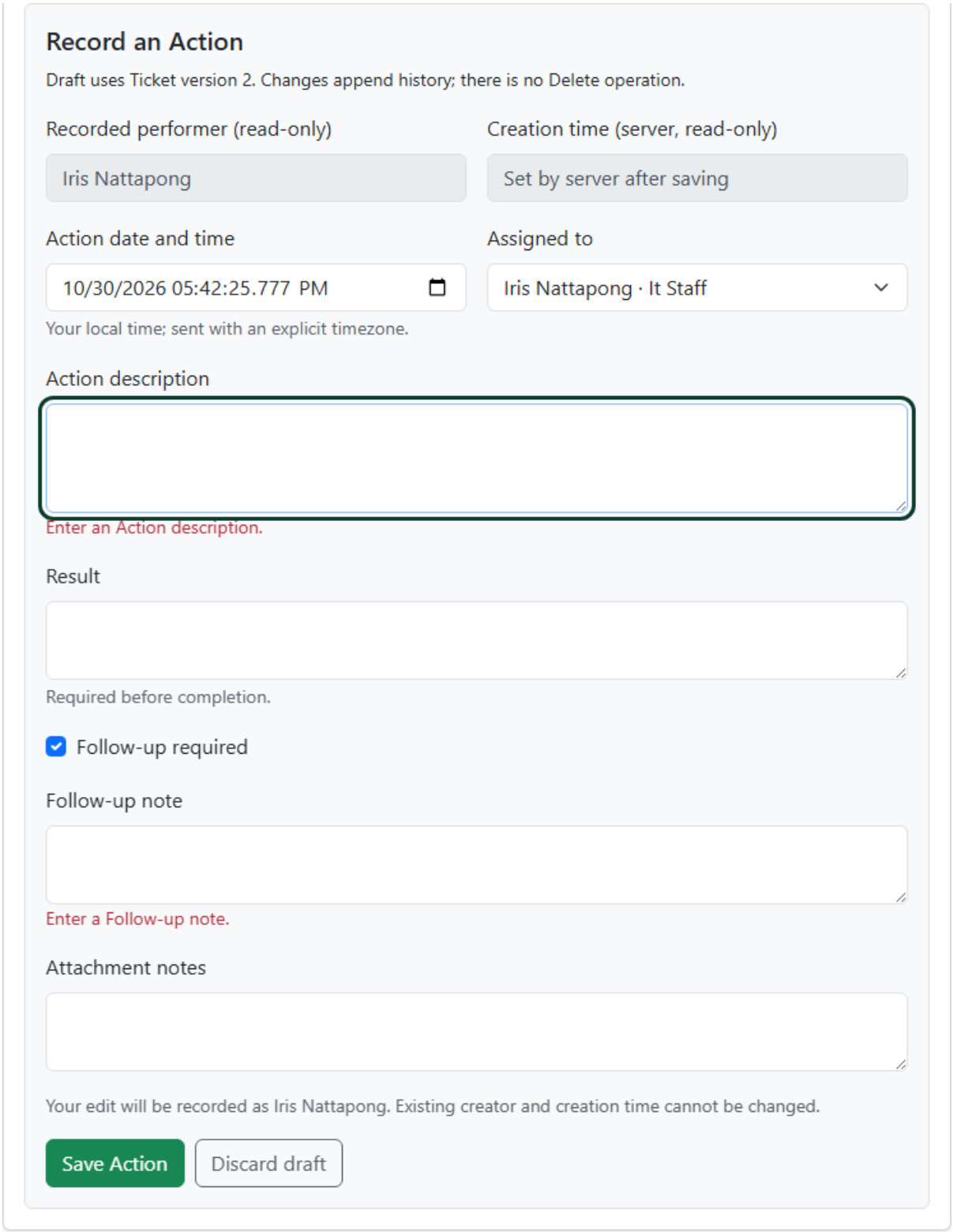


Figure 11. Actual labelled Action form: near-field required/conditional validation; actor is read-only. Focused complete Record an Action form; preceding saved card omitted. Full original

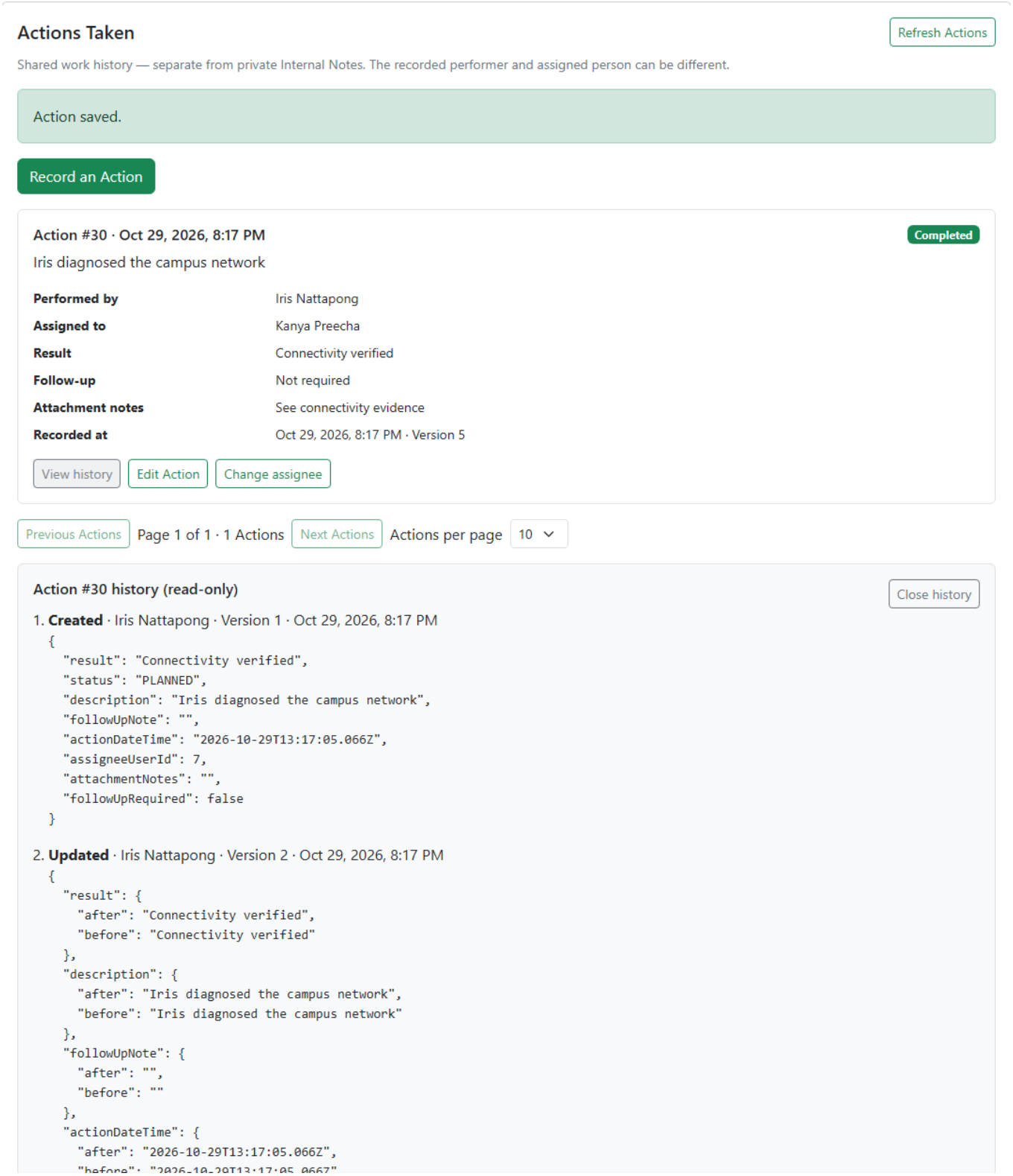


Figure 12 (1/2). Actual Action lifecycle with completion/cancellation and append-only change events; history remains visible. Full original

```
    },
    "attachmentNotes": {
      "after": "See connectivity evidence",
      "before": ""
    },
    "followUpRequired": {
      "after": false,
      "before": false
    }
  }
```


3. **Assigned** · Iris Nattapong · Version 3 · Oct 29, 2026, 8:17 PM


```
  {
    "assigneeUserId": {
      "after": 8,
      "before": 7
    }
  }
```


4. **Status Changed** · Iris Nattapong · Version 4 · Oct 29, 2026, 8:17 PM


```
  {
    "status": {
      "after": "IN_PROGRESS",
      "before": "PLANNED"
    }
  }
```


5. **Status Changed** · Iris Nattapong · Version 5 · Oct 29, 2026, 8:17 PM


```
  {
    "status": {
      "after": "COMPLETED",
      "before": "IN_PROGRESS"
    }
  }
```


Figure 12 (2/2). Contiguous continuation of the same original, no content reconstructed. Full original

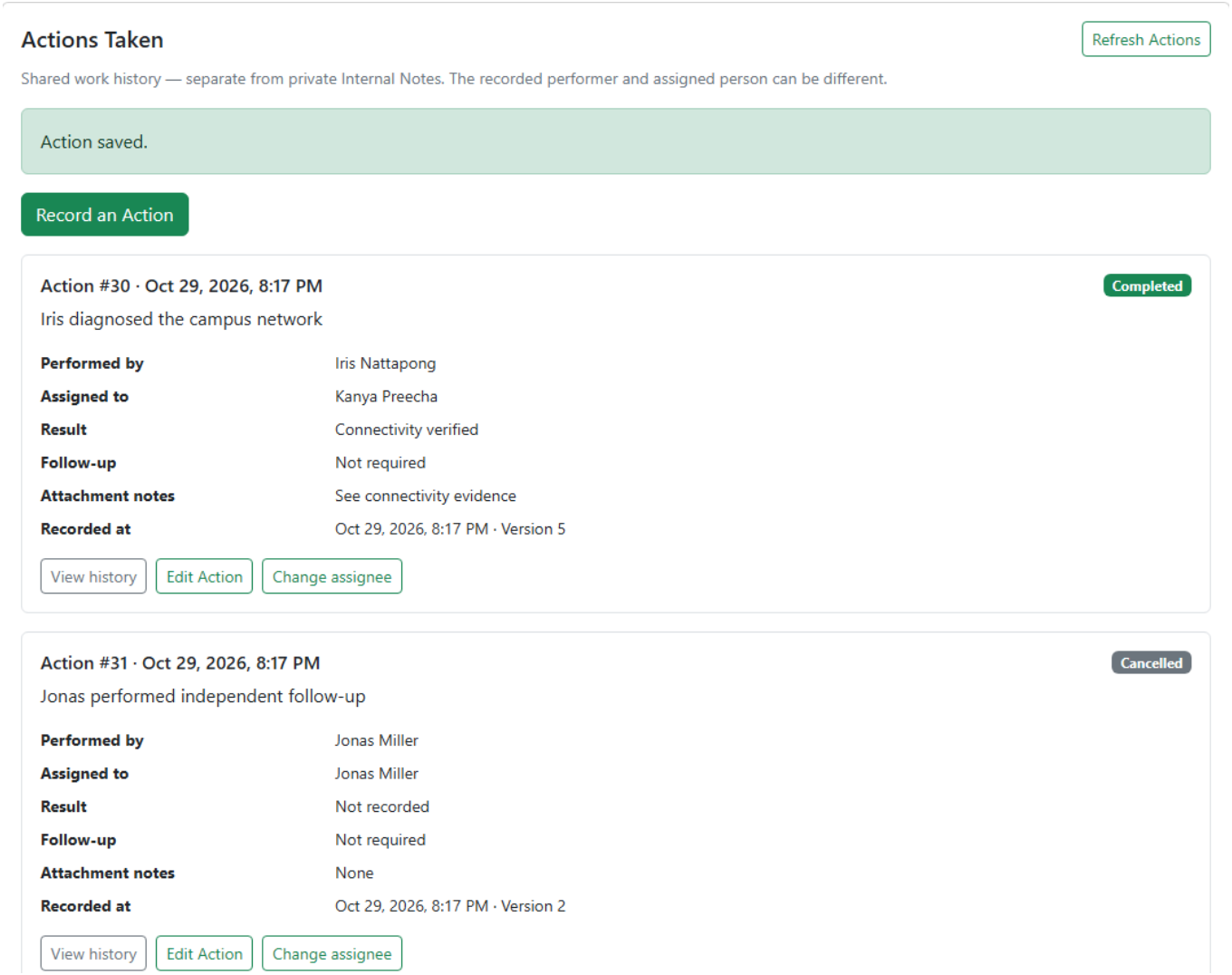


Figure 13. Real multiple staff working on one Ticket, separate actor and assignee; not limited to Ticket Owner. Top viewport only; full original linked. Full original

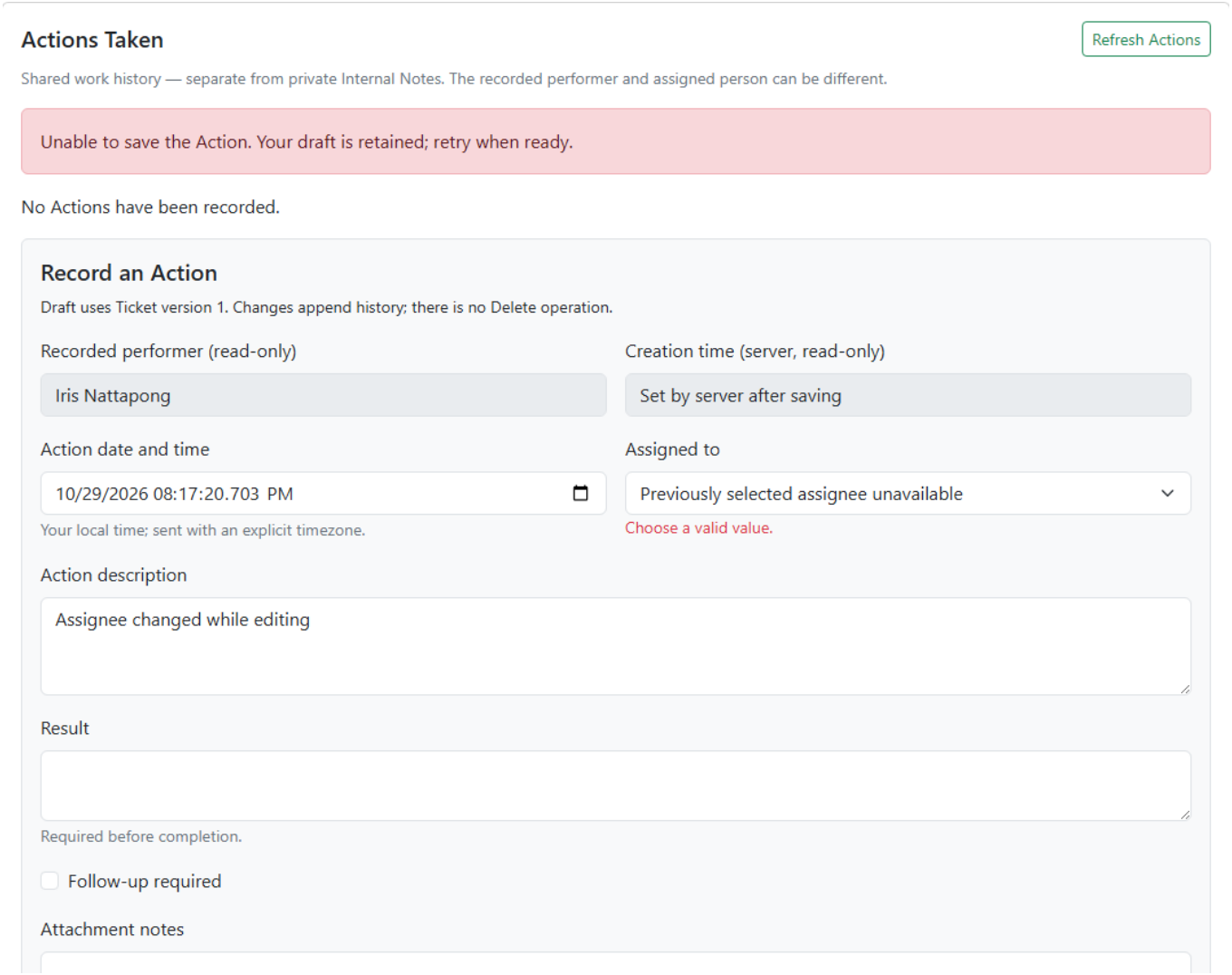


Figure 14. Actual inactive-assignee validation; input retained and no unsafe assignment saved. Top viewport only; full original linked. Full original

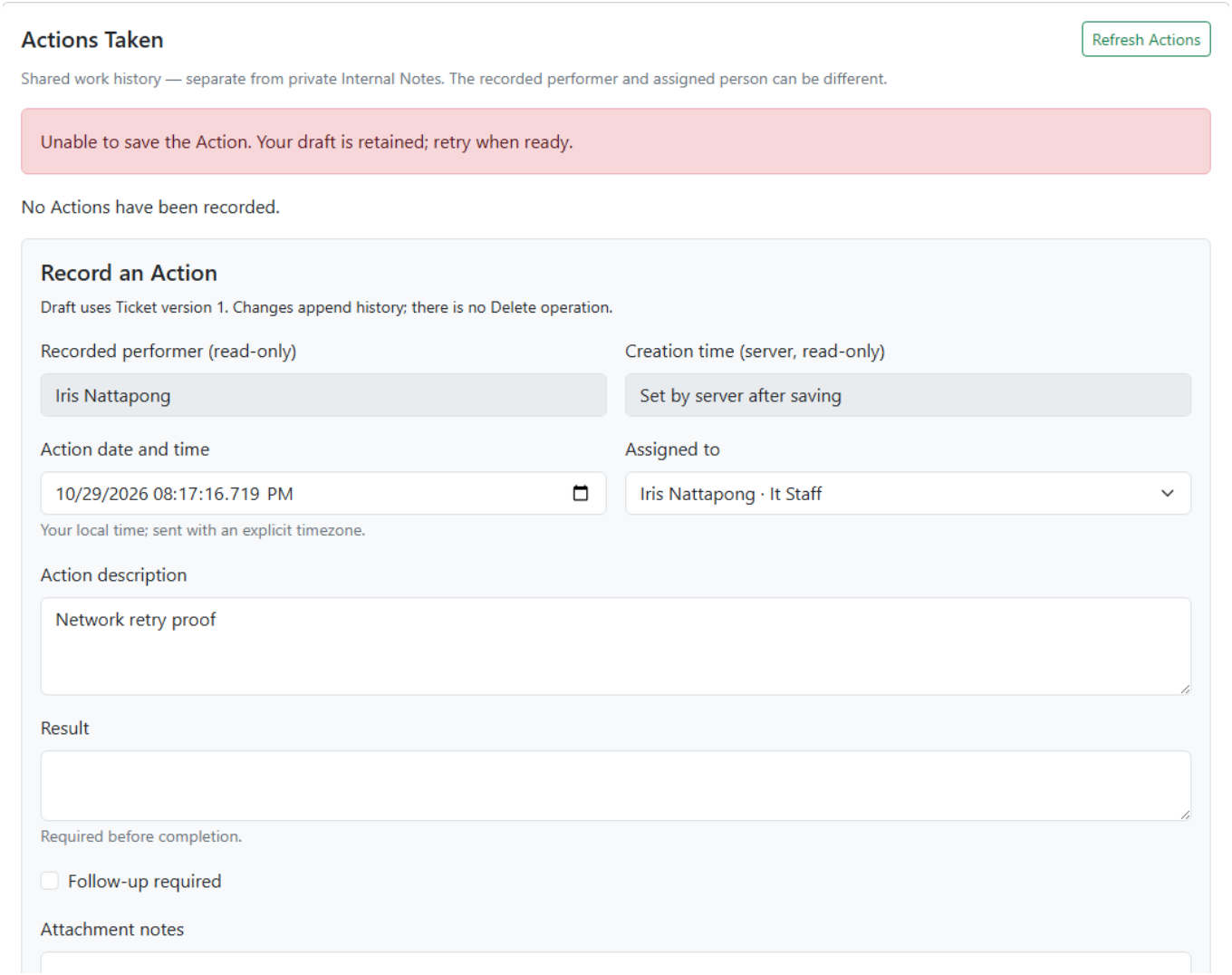


Figure 15. Controlled lost-response recovery: durable logical retry key prevents a second Action. Injection explicitly labelled. Top viewport only; full original linked. Full original

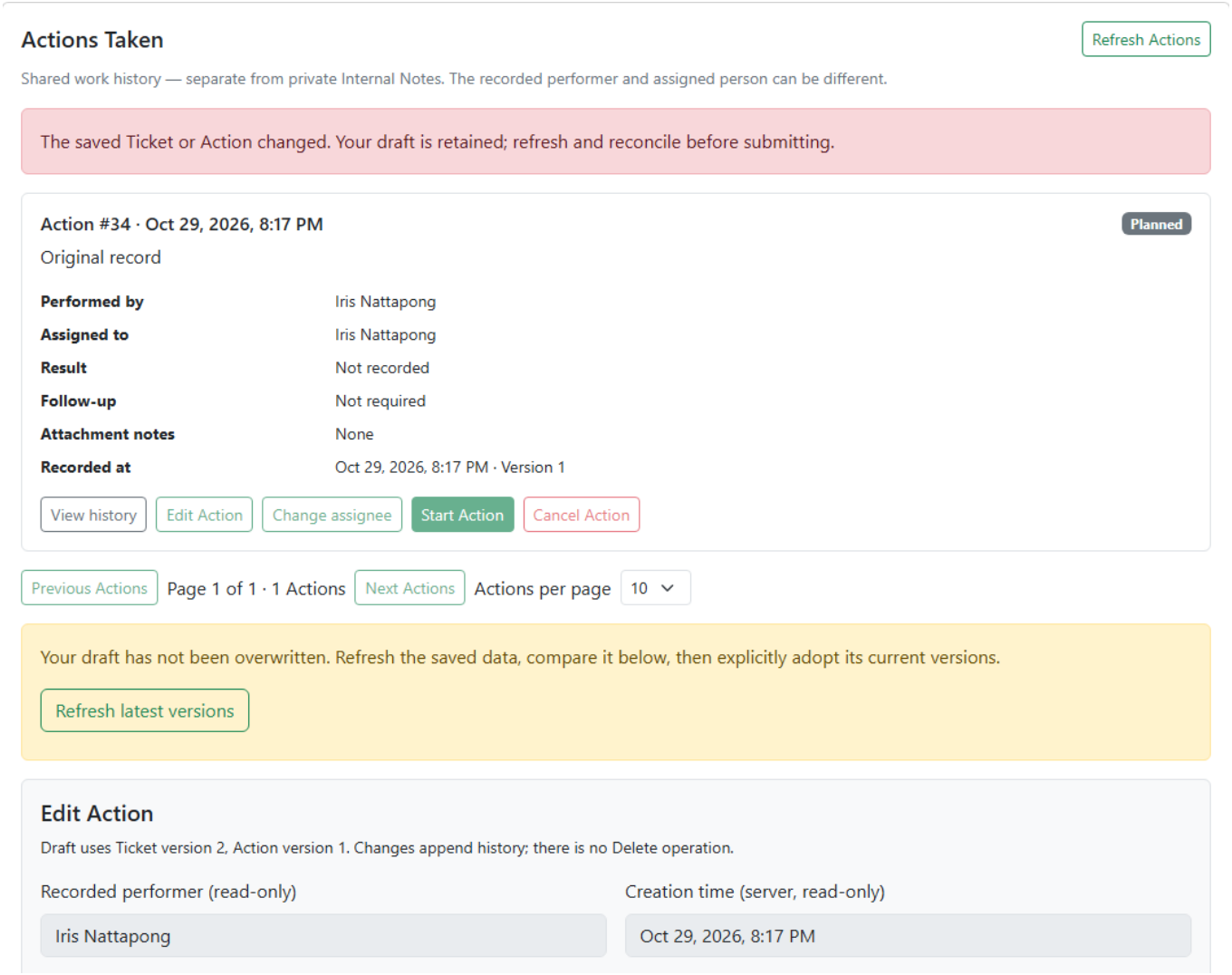


Figure 16. Real stale-version conflict: preserve draft and require deliberate refresh/reconciliation, never silently overwrite. Top viewport only; full original linked. Full original

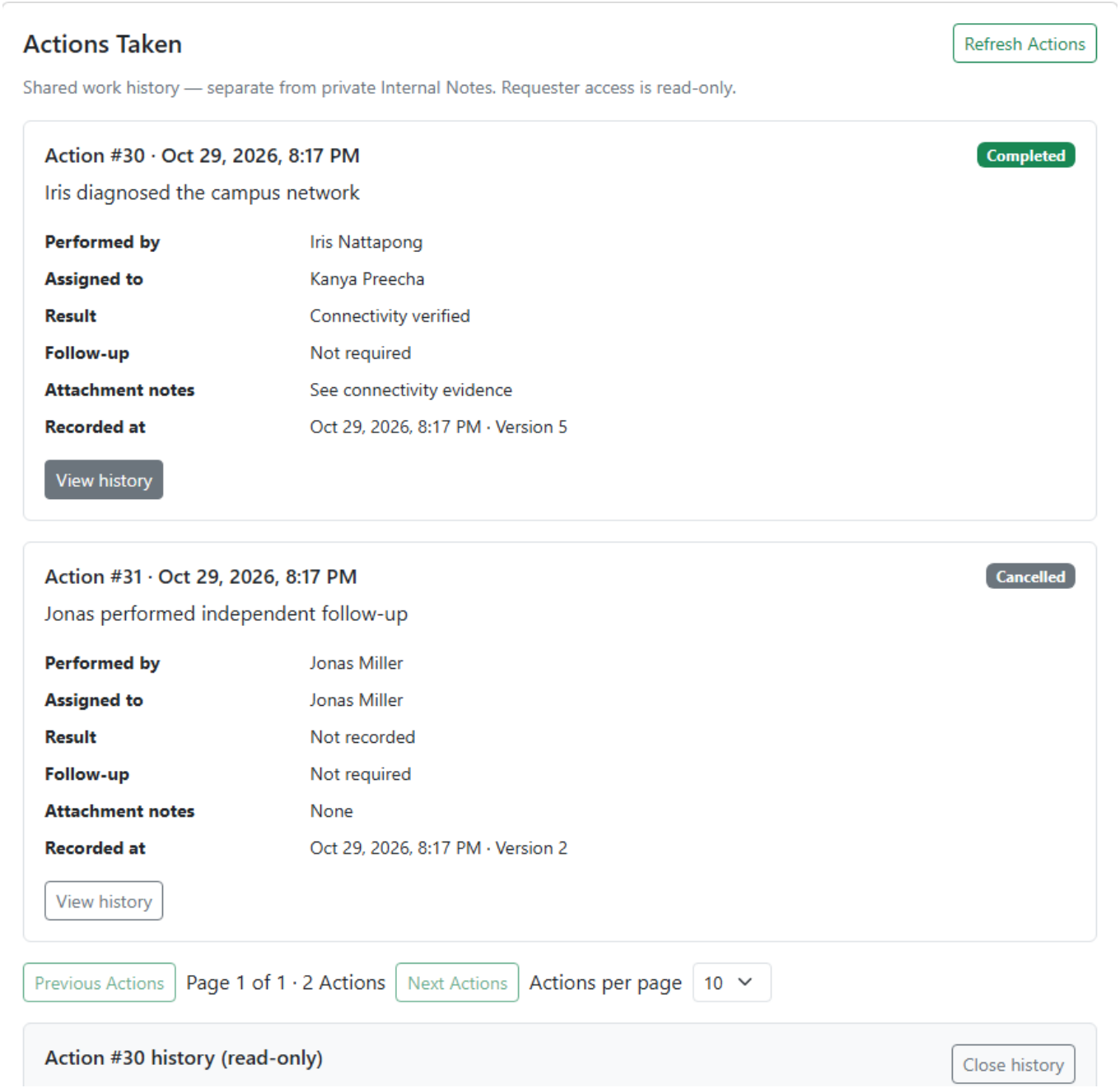


Figure 17. Owned Requester sees shared Actions read-only and no Staff mutation controls. Top viewport only; full original linked. Full original

Real API tests cover spoofed actor, wrong/inactive assignee, invalid time/text, cross-owner404, wrong-role403, terminal state rejection, same-key replay/different-payload409, event ordering, rollback and races. See the complete output in Part3; disabled UI buttons alone are not authorization proof.

Answer Part 7 - Working Ticket Workflow

The approved eight-status matrix is preserved. Resolve requires at least one Completed Action, no Planned/In-Progress Action and no outstanding follow-up. Cancellation requires unfinished Actions to be explicitly cancelled first. Requester indication stays advisory. Staff/Admin formal transitions are versioned, transactional and recorded in immutable ordered history.

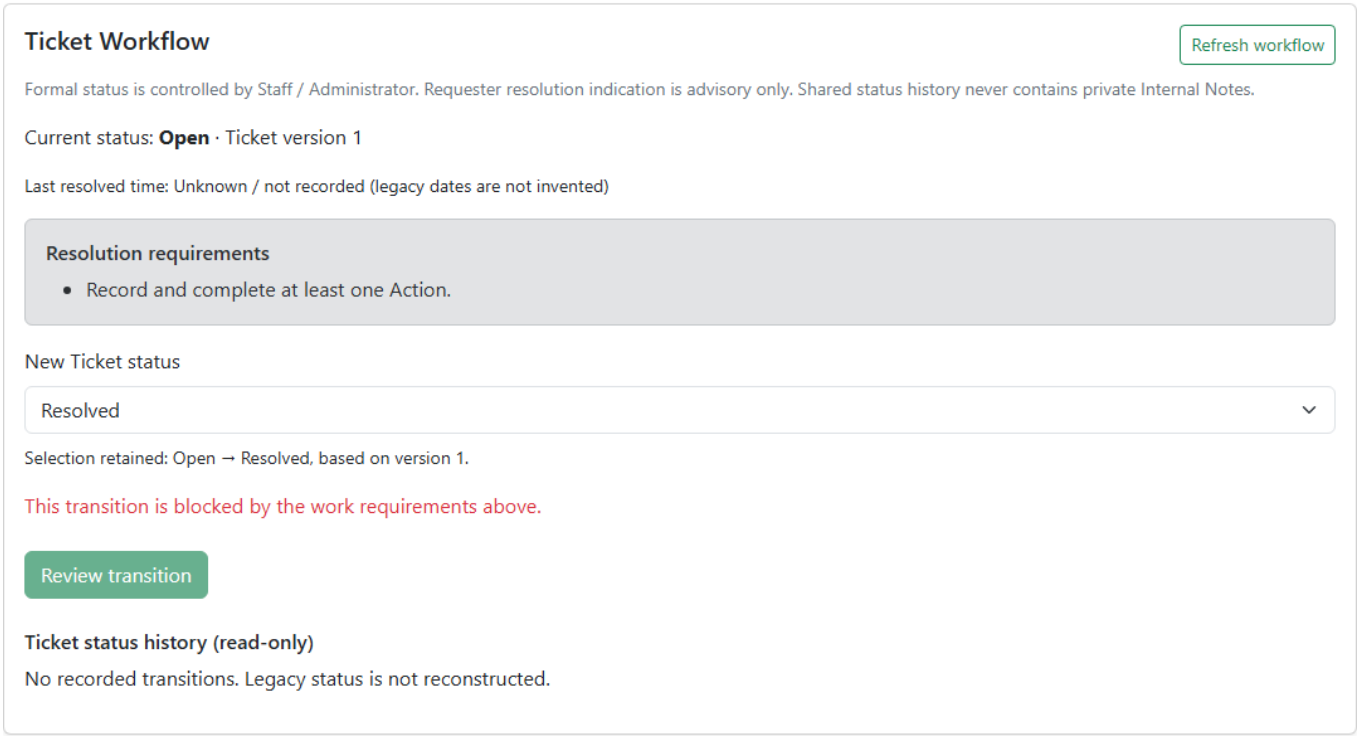


Figure 18. Real gate: no Completed work blocks Resolve. Full original

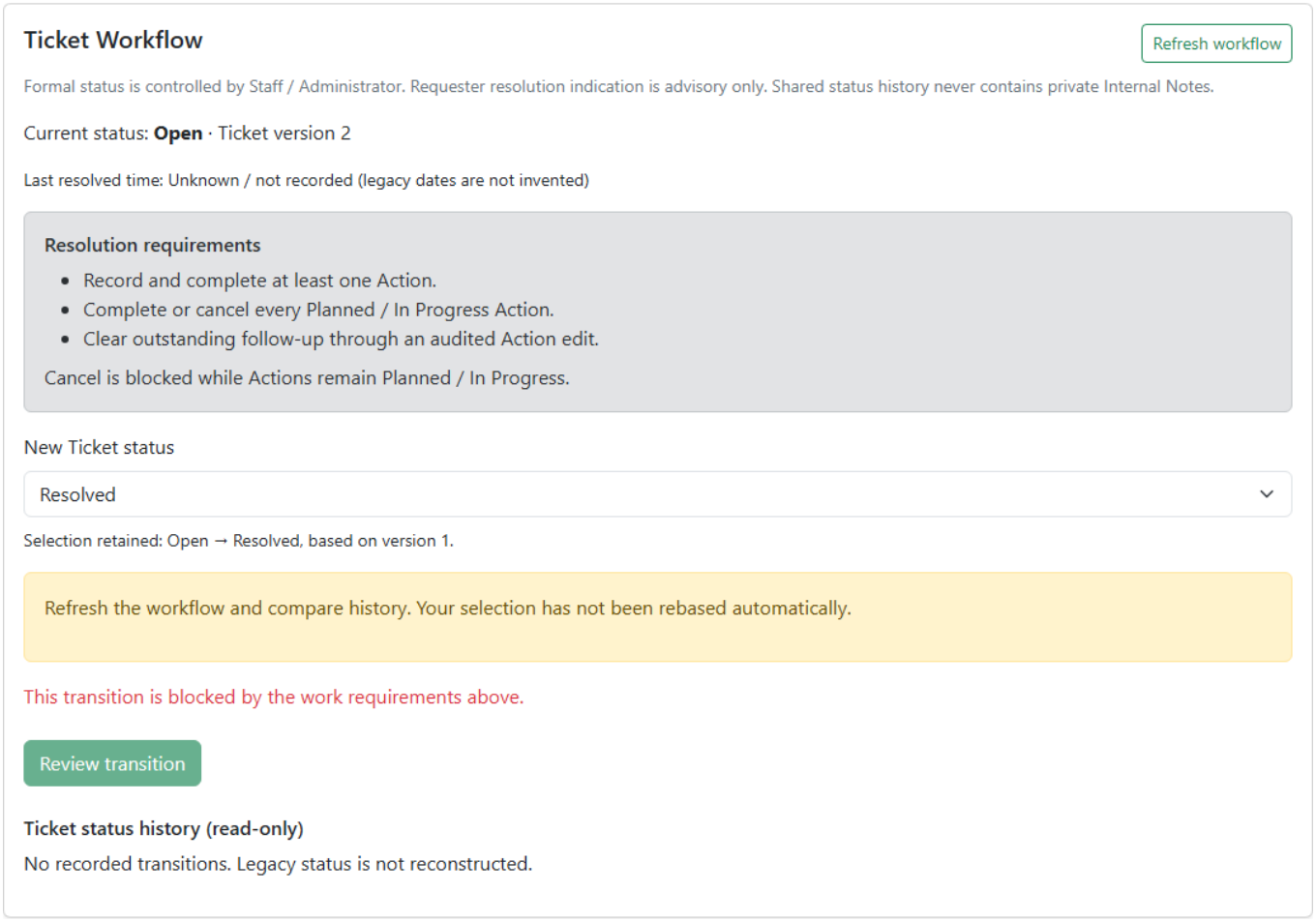


Figure 19. Real gate: unfinished Action blocks Resolve even when another Action is Completed. Full original

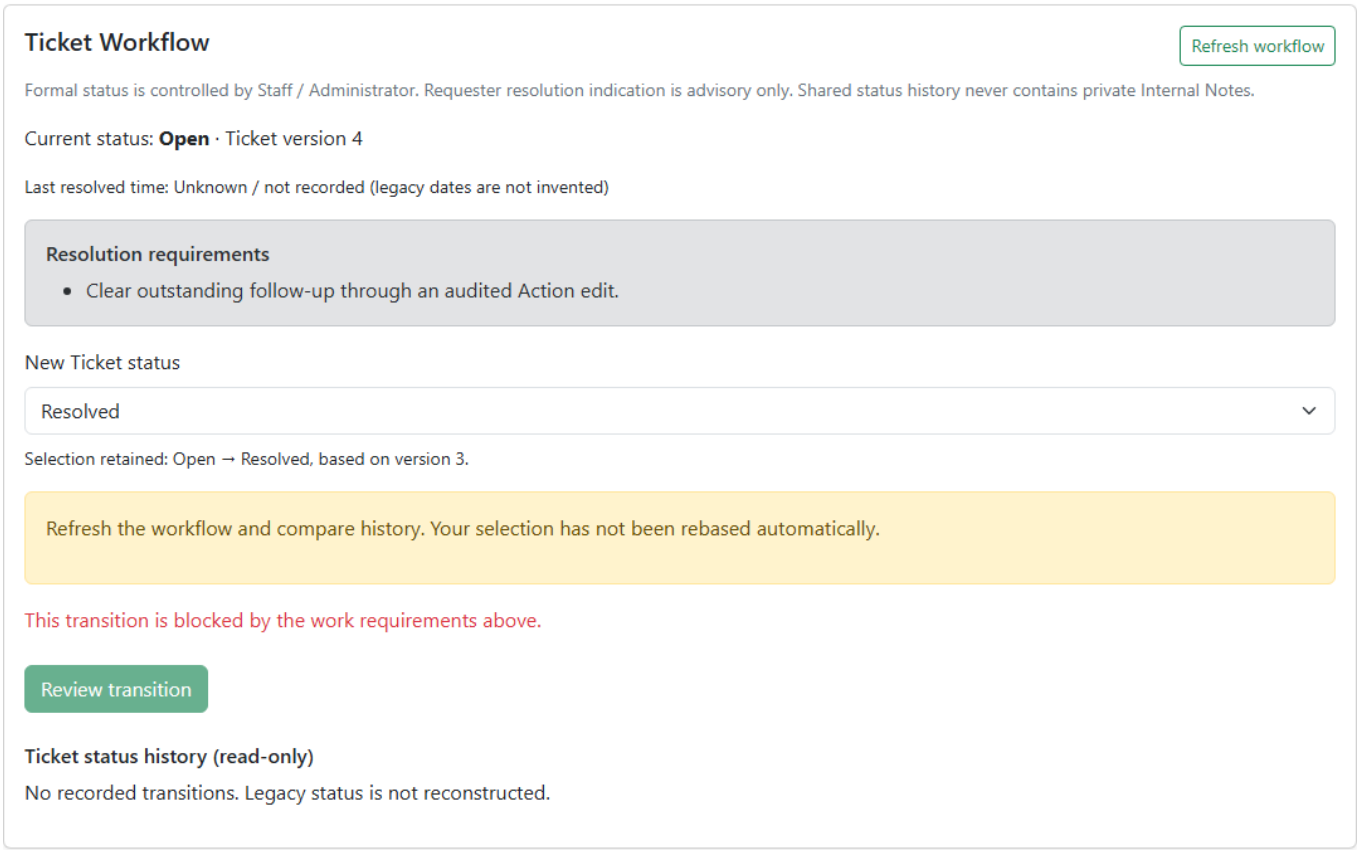


Figure 20. Real gate: outstanding follow-up blocks Resolve. Full original

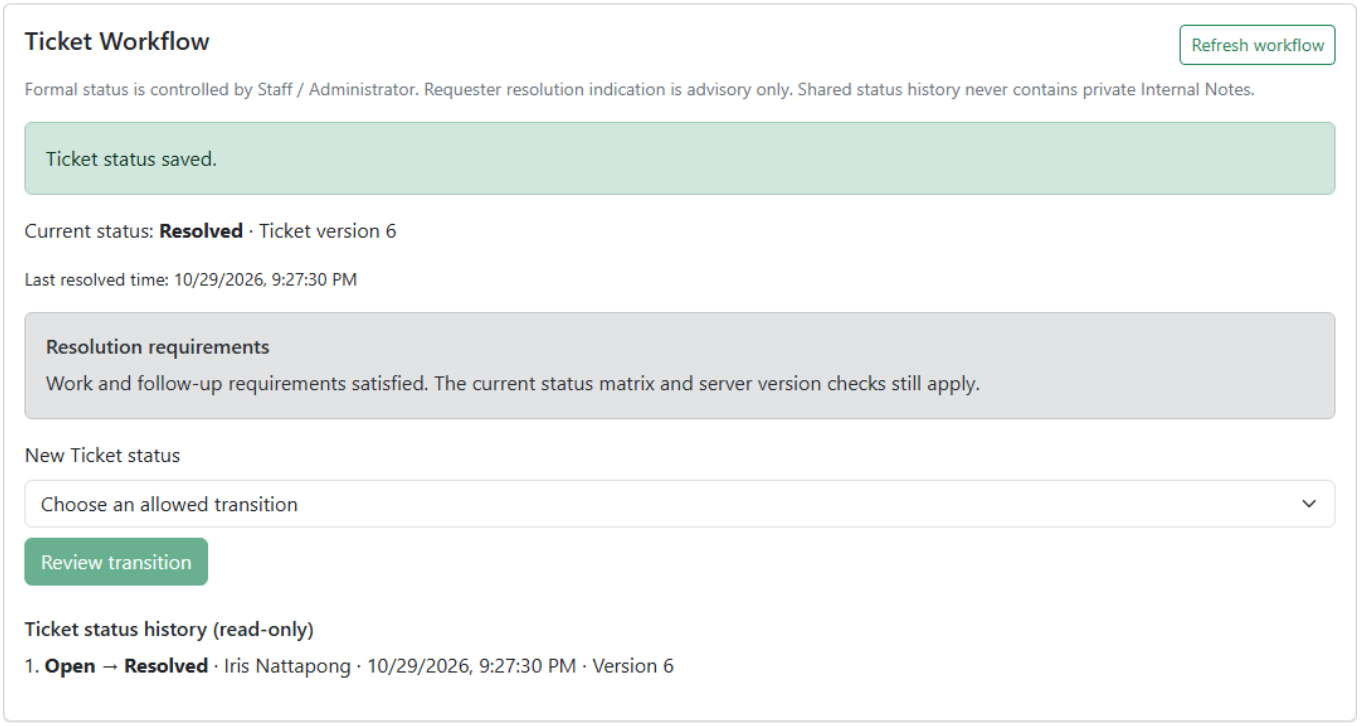


Figure 21. Actual successful Resolve after satisfying all predicates, with recorded actor/time/history. Full original

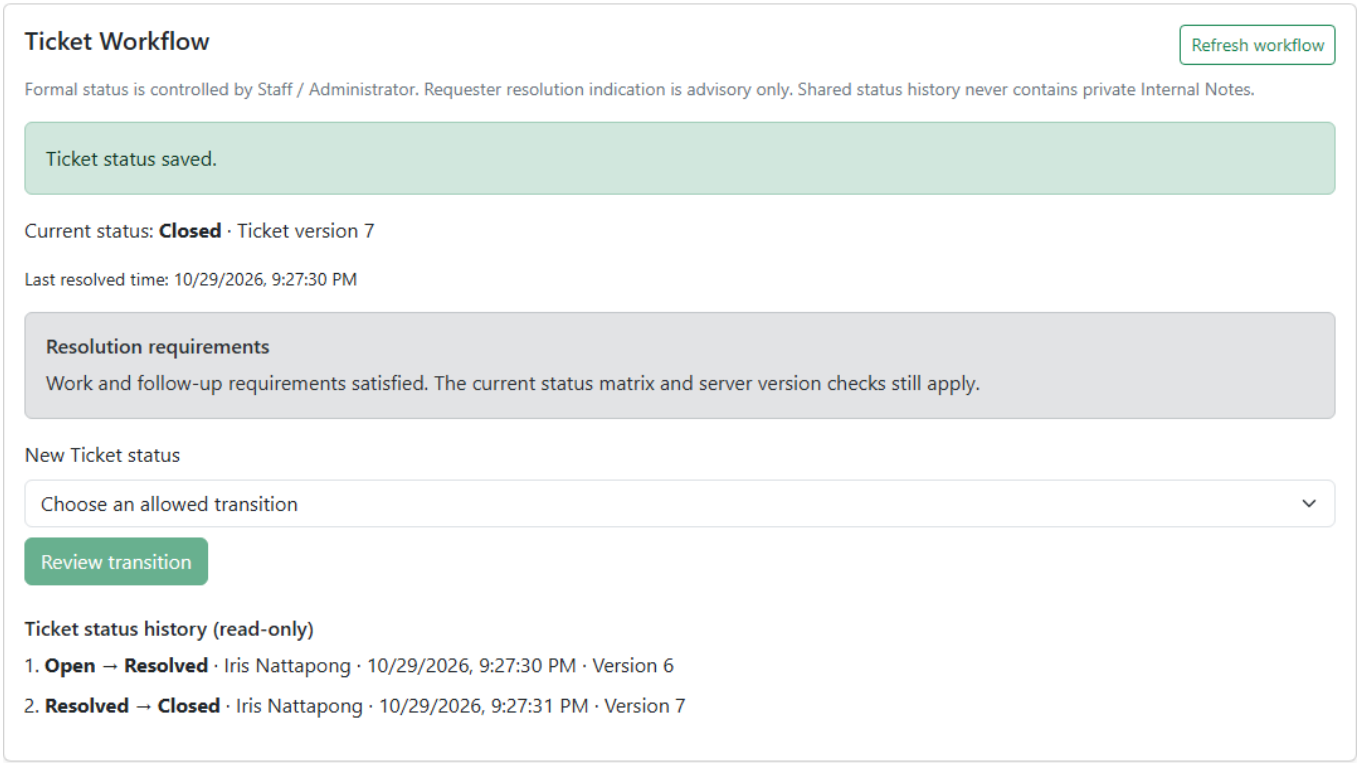


Figure 22. Actual Resolved-to-Closed transition and ordered history. Full original

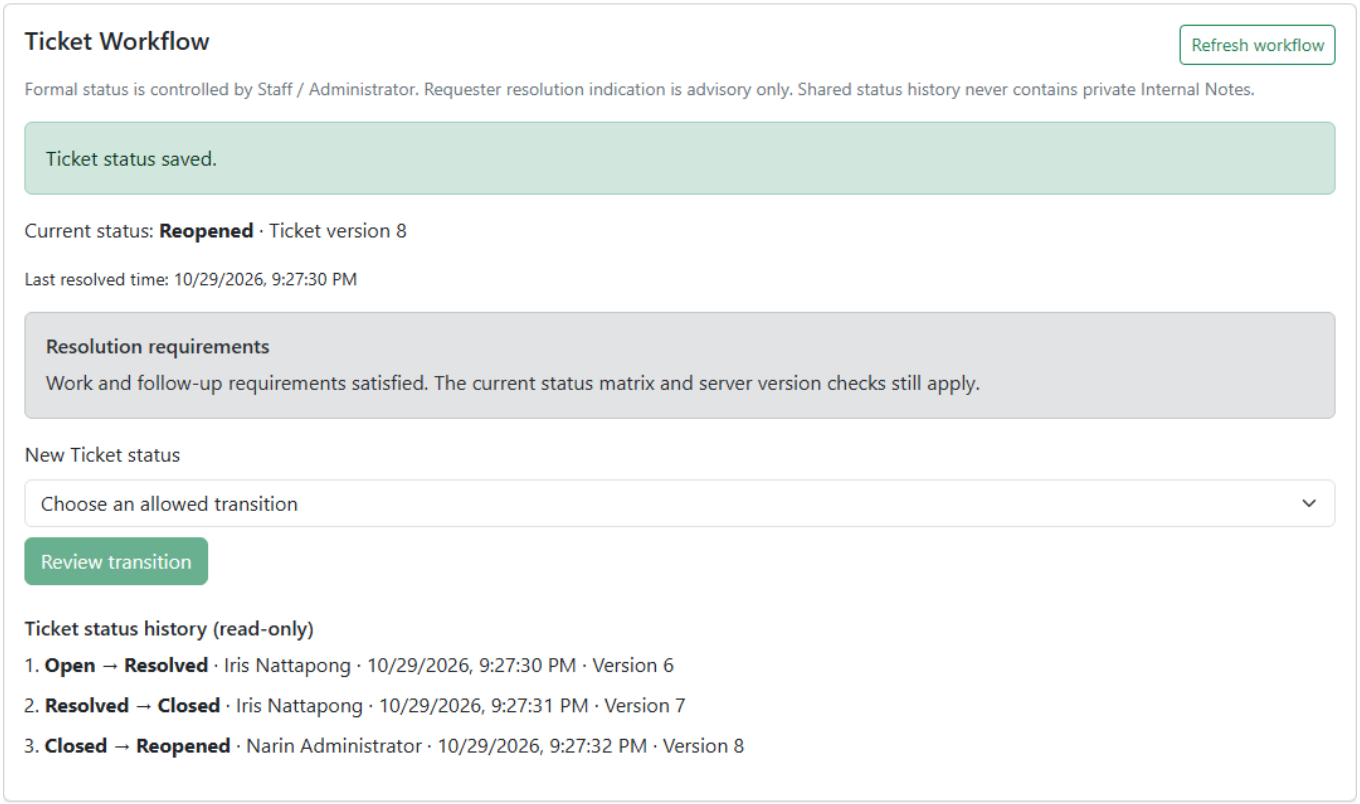


Figure 23. Actual Administrator reopen with append-only prior history preserved. Full original

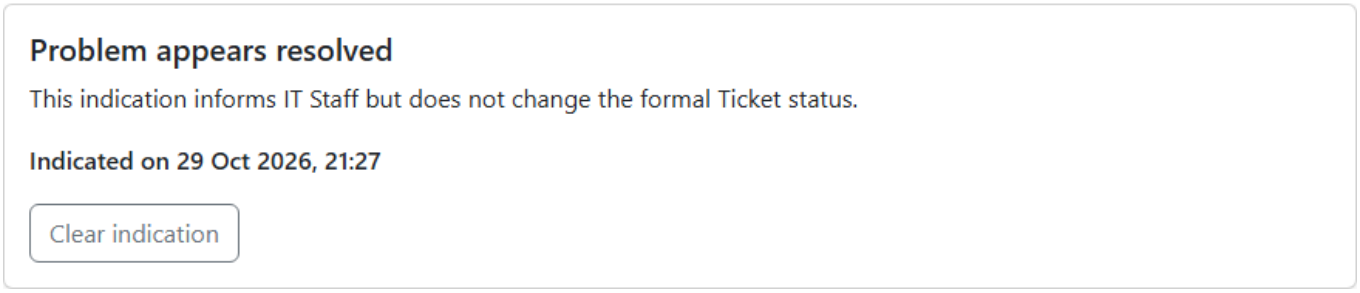


Figure 24. Requester indication is visible but does not perform a formal Ticket transition. Full original

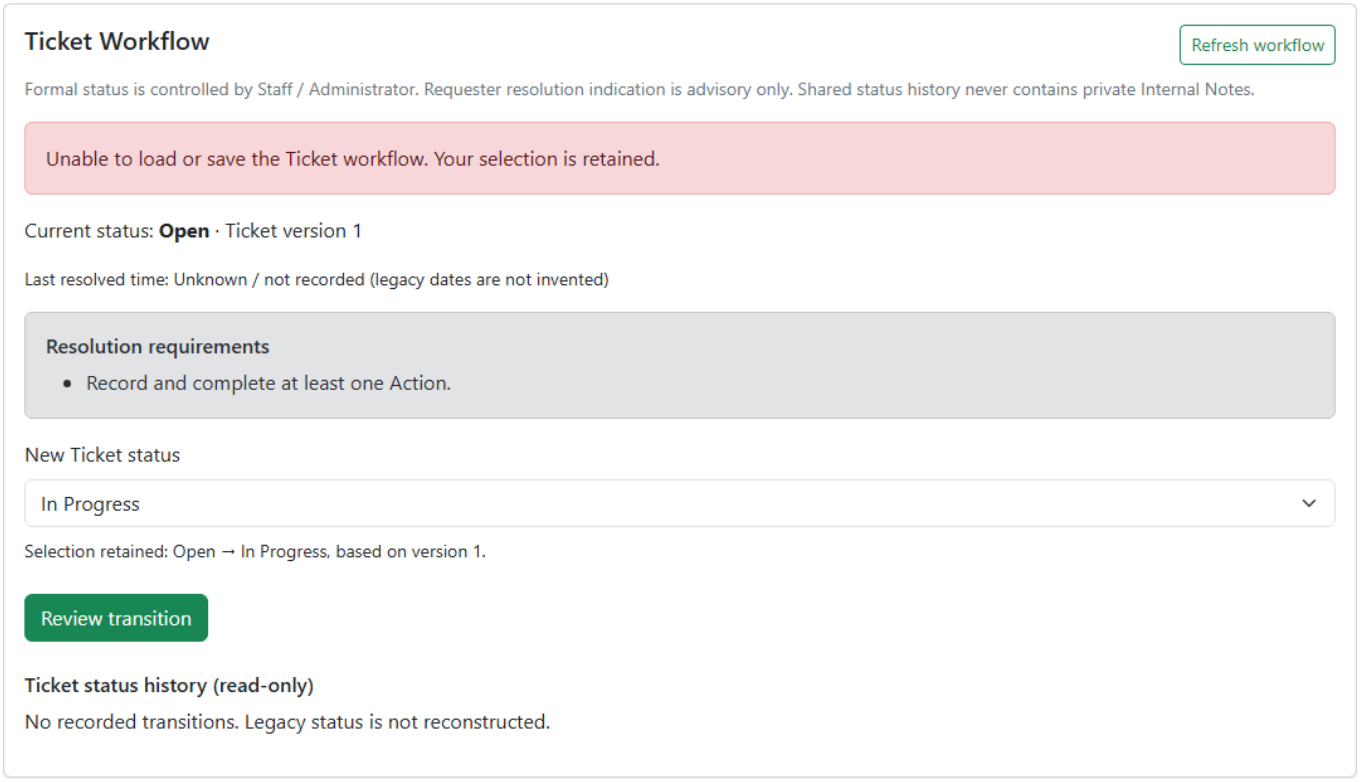


Figure 25. Explicit injected save failure retains the intended transition for safe recovery. Full original

All8x8 transition pairs are exercised for permitted roles; unlisted transitions, stale versions and Action-vs-Resolve races are rejected. History cannot be overwritten/deleted through application routes. Full event/negative assertions are in Part3, not inferred solely from the screenshots.

Answer Part 8 - Working Requester Dashboard and Final Regression UI

Requester metrics use session-owned Tickets: Open, Waiting for Requester and recently resolved within the defined Bangkok calendar window, plus attention and recents. Last-resolved legacy nulls remain unknown, not invented. Card filters match the destination list; refresh obtains a new snapshot.

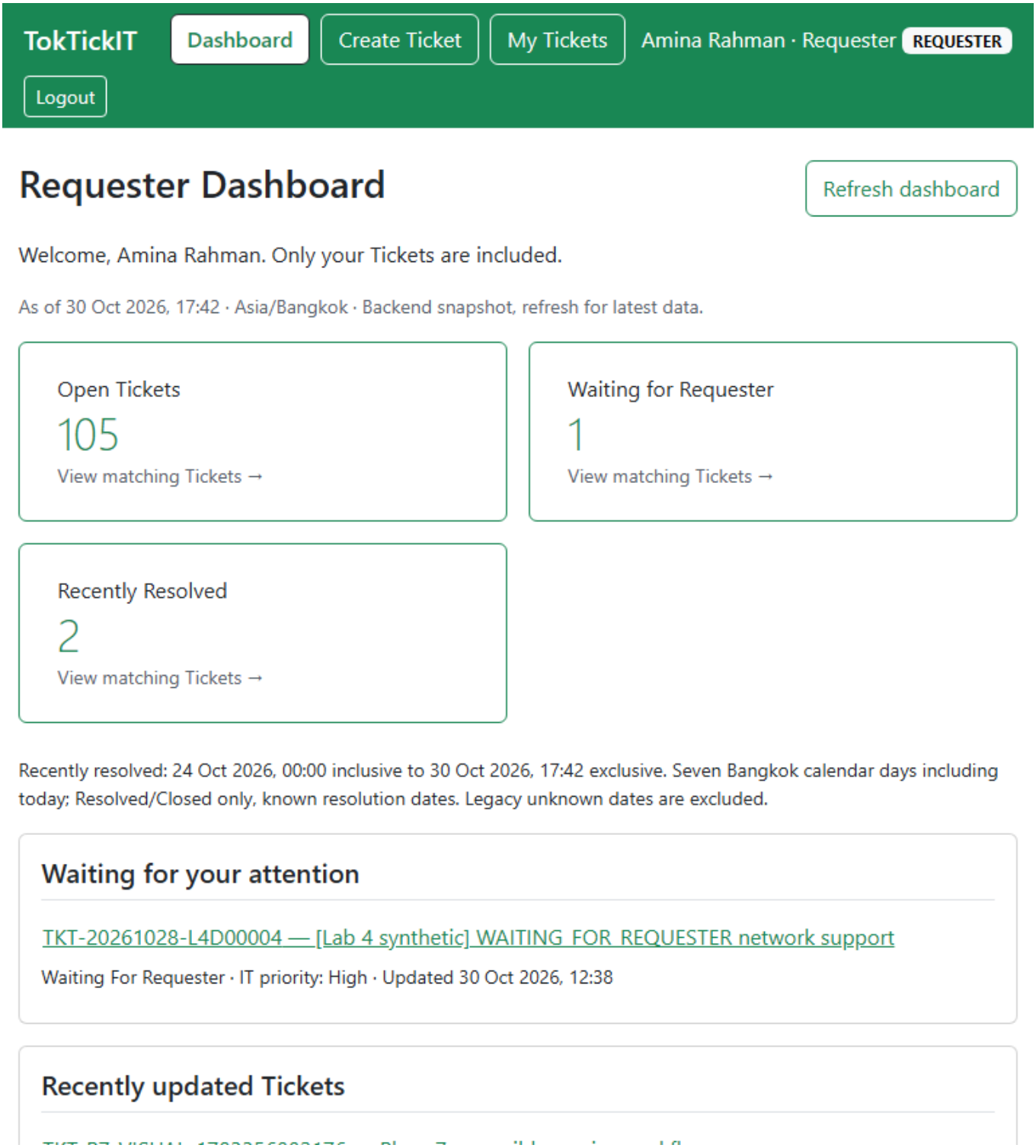


Figure 26. Actual owned Requester Dashboard: counts, attention and recent items with detail/list destinations. Top viewport only; full original linked. Full original

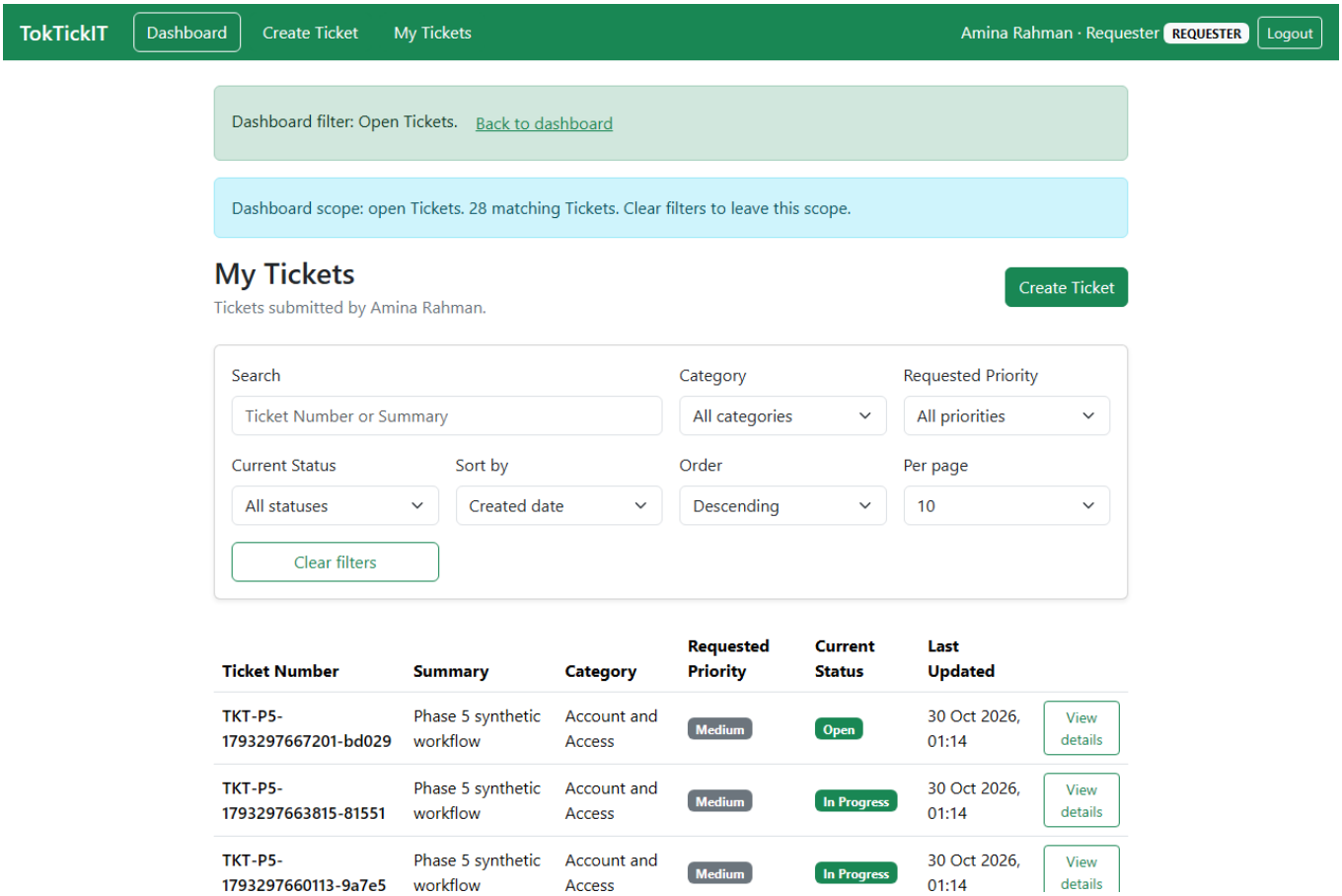


Figure 27. Real card drill-down with matching owned-list predicate and clearable source context. Top viewport only; full original linked. Full original

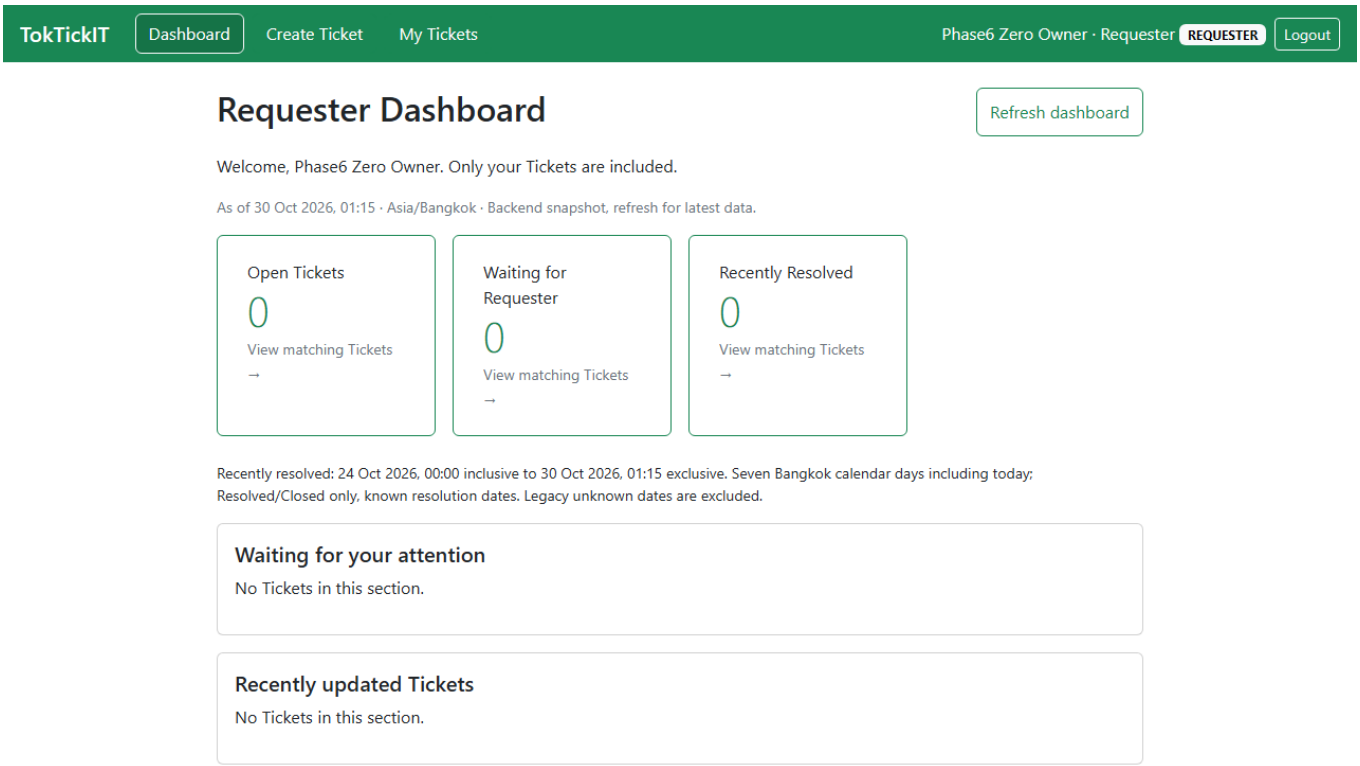


Figure 28. Real new synthetic Requester: numeric zero counts and empty lists; not an injected success payload. Full original

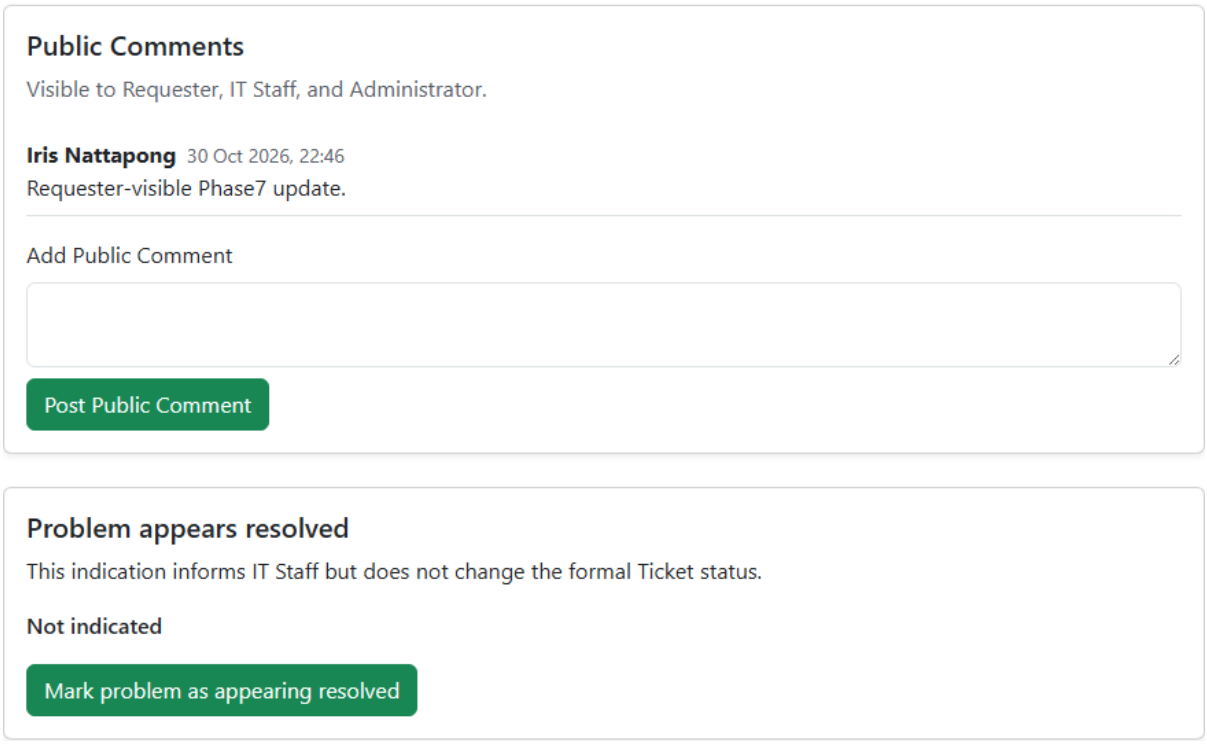


Figure 29. Current Lab4 regression: owned Public Comment and advisory indication. Focused shared panels; private-Note absence is checked across the full DOM/API, not inferred from this crop alone. Contiguous Public Comments/indication region; Ticket identity above retained in full original. Full original

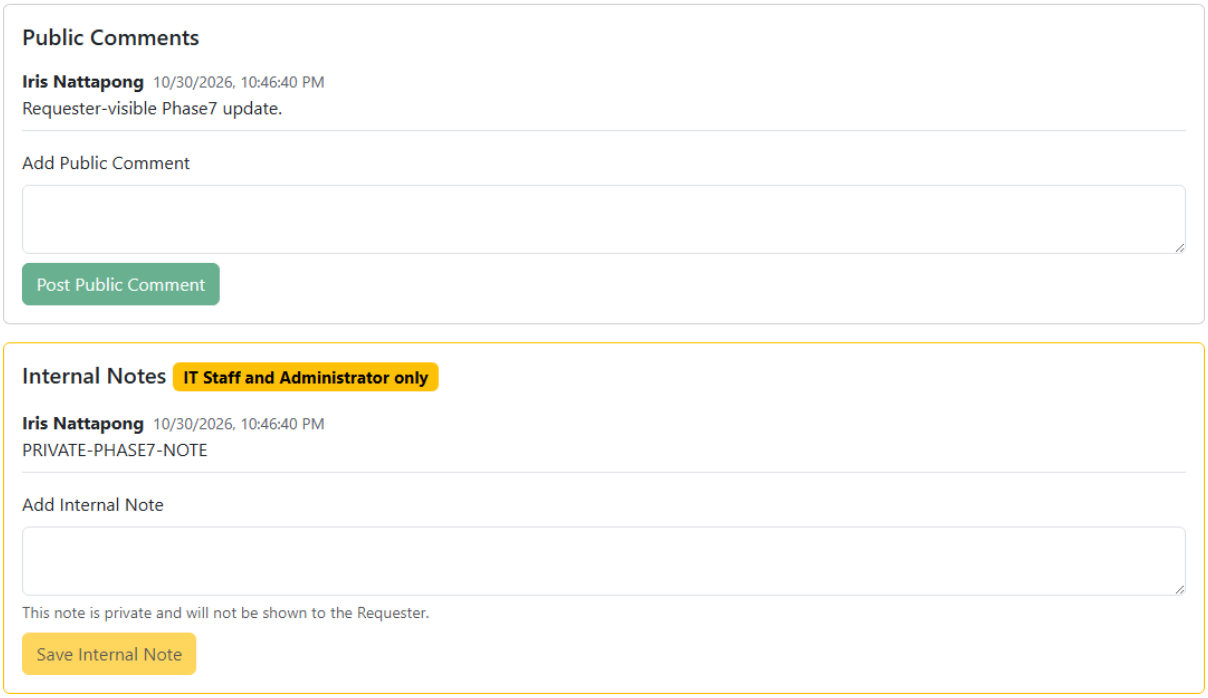


Figure 30. Current Lab4 Staff regression: Public Comments and clearly separated private Internal Notes. Contiguous Public Comments/private Notes region; upper Ticket controls omitted. Full original

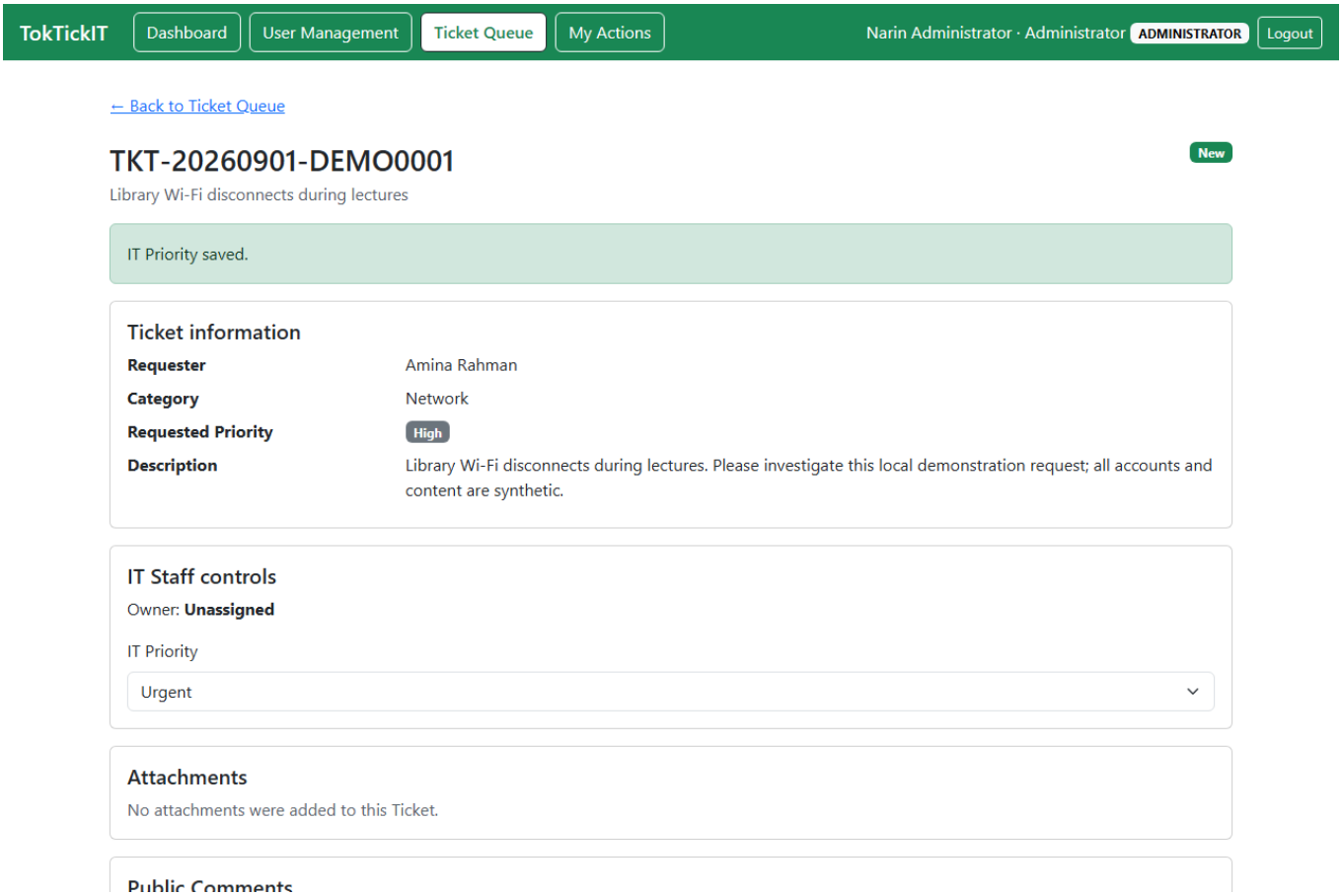


Figure 31. Current Lab4 Administrator regression: permitted IT Priority correction without Staff-only owner/comment/Note operations. Top viewport only; full original linked. Full original

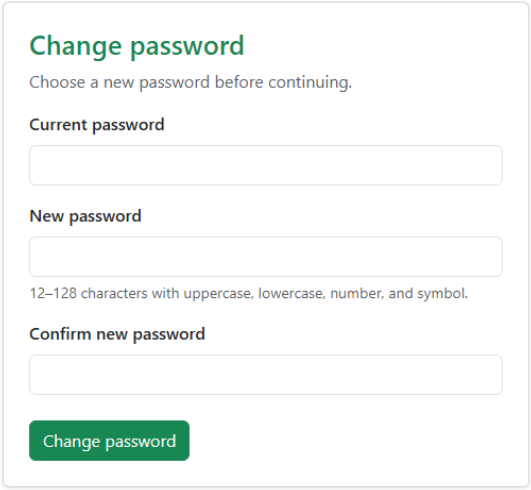


Figure 32. Preserved first-login password-change gate; existing authentication/role regression remains executable. Exterior blank margins omitted only. Full original

The complete37-scenario browser run preserves Lab2 create/search/detail and Attachment upload/removal/blocked-download404, plus Lab3 login/password, Staff/Admin and private-Note boundaries. Actual source assertions, not new screenshots of every unchanged baseline operation, establish regression coverage. Final-main execution is still pending release; this heading follows the handout and does not mean the preview itself is a final release.

Answer Part 9 - Zen Green UI, Responsive, Accessibility, and Final Polish

Current reusable UI specification and completed, qualified inspection follow. Desktop1440x900, tablet768x1024 and mobile390x844 full originals are retained. The following viewport extracts establish layout; earlier sections show readable functional panels. Long content below each extract is intentionally omitted here, never represented as a full-page screenshot. Full originals are linked, and whole-document overflow assertions cover the long screens.

Rendered docs/lab-04/ui-spec.md

Exact source / complete raw output

Lab 4 UI Specification and Inspection

Current30October2026: product Phases1-7 accepted at staging b887994. Historical notes preserve earlier implementation states. This checklist records actual local inspection, not final release/main/PDF QA.

Reuse existing Zen Green conventions; no replacement design or new shell framework. Requester Dashboard and Staff/Admin Dashboard lead to detailed lists/Tickets. Actions area belongs to existing Ticket Detail: list plus create/view/edit modes, assignment and explicit status controls. Requesters have read-only own-Ticket work.

Display date/time, description, result, authenticated performed-by, separate assignee, follow-up indicator/note and attachment notes. Preserve terminal Actions/history. UI transitions reflect backend eligibility but never replace backend authorization. Show a resolution explanation when the approved gate is unmet; Requester indication is advisory. Distinguish shared Actions from private InternalNotes without color alone.

Every screen needs loading, success, zero/empty vs no-results, validation, forbidden, not-found, stale-conflict and API-failure feedback. Field errors are near fields and semantically associated. Busy submits prevent duplicate clicks. Recoverable failures retain input; conflicts require deliberate refresh/reconciliation, not silent overwrite.

Verified inspection viewports: desktop 1440x900, tablet 768x1024, mobile 390x844. Inspect all major screens at all three; additionally keyboard traversal, visible focus, semantic labels/headings, non-color cues, readable contrast, dialogs, clipping, overlap and horizontal overflow. Screenshot only fully loaded readable content.

Visual/accessibility check	Current status
Zen Green design consistency and role navigation	Implemented; current nav semantics and skip link verified in Phase7
Dashboard cards, current-user Actions and drill-down	Actual DB/API/component/browser evidence in Phase6/7
Action list/forms/status/read-only visibility	Loaded desktop/tablet/mobile captures and role tests
Editable/read-only fields and validation placement	Gray read-only fields, near-field text/aria errors; three-width inspection
Button hierarchy, loading and recoverable feedback	Primary/secondary controls; retained drafts/retries; confirmed writes clear input
Keyboard/focus/semantic labels/non-color cues/contrast	Selected keyboard paths and axe automated scans pass after measured contrast fixes; not full assistive-tech certification
Clipping, overlap and horizontal overflow	Loaded screenshots inspected, document-width assertions pass; normal vertical scroll

No checkbox is marked complete before actual inspection.

Desktop: up to four-column metric grid, readable tables/recent lists; tablet two columns, mobile one-column cards and labelled stacked rows. Exact metric drill-down links preserve role filters; My Actions paginated list links staff Ticket Detail.

Action form contains local datetime converted to ISO+timezone, description/result, active assignee, follow-up checkbox/conditional note and attachment notes. Actor and createdAt read-only. Create starts Planned; terminal status cannot reopen, permitted annotations still append history on active Tickets. Requester has no write controls. One idempotency key per logical create, retained with input through retry/failure. 409 shows reload/reconcile, never overwrites automatically. Success refreshes Ticket version/status, Action/history and dashboard. Semantic labels, keyboard focus, textual statuses/priorities and field-associated errors required at all viewports.

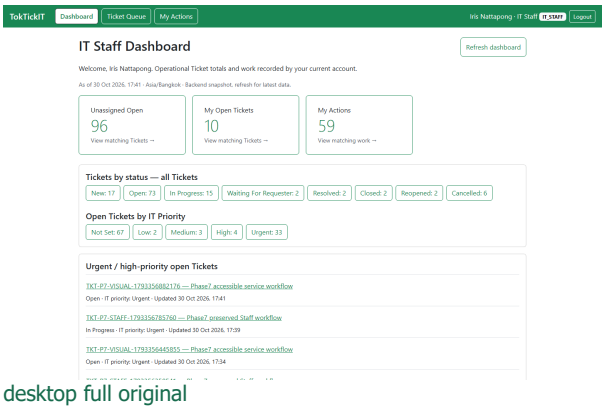
Navigation, data refresh and evidence limits

Explicit Dashboard navigation preserves legacy landing pages. One current nav destination, skip link and visible focus are semantic, not color-only cues. Dashboard filters persist into list search/sort/pagination; source context is clearable. Back from Detail returns to its source. Remount/Refresh re-fetch; errors clear stale metrics. My Actions means performedBy, not assignments. Requester/shared Actions and private Internal Notes remain distinct and backend protected.

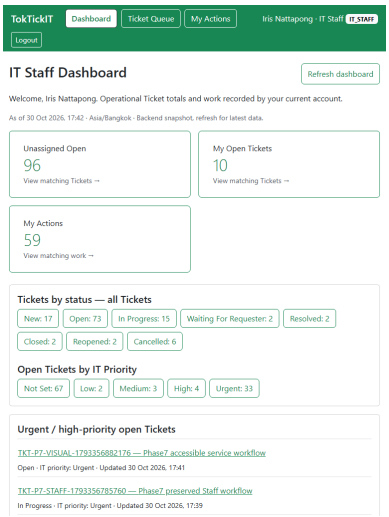
Phase7:64 inspected images,48 automatic axe reports with zero violations and10 console records with zero unexpected errors. Exact expected unauthenticated auth/me401 and injected Note503 retained. Native select SVG backgrounds make some contrast checks incomplete; actual text color pairs checked manually. Not screen-reader or WCAG certification. Phase8 Staff-specific delayed request, injected503/403, real zero-personal-work and retry captures supplement earlier Requester state images. All injections explicitly labelled; real API authorization tests remain separate.

Color fixes preserve Zen Green: navbar pure white on #198754 is4.53:1; validation #b02a37 on #f8f9fa is6.16:1; read-only #495057/#e9ecef is6.90:1. Role badges have a supported accessible name, pagination a navigation landmark. Source facts and loaded screenshot inventory: phase7-visual-evidence.md.

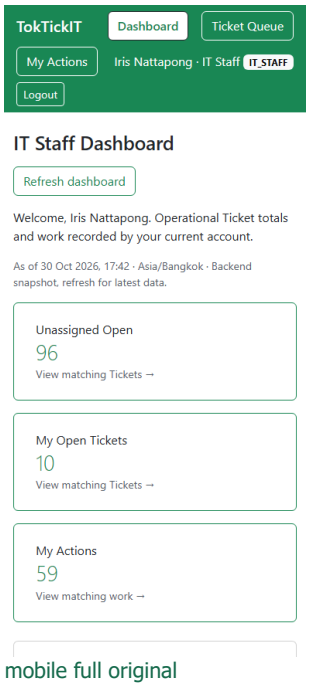
Staff Dashboard - three-width layout comparison



desktop full original



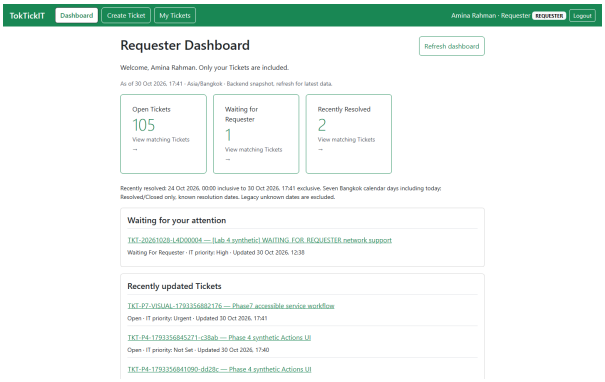
tablet full original



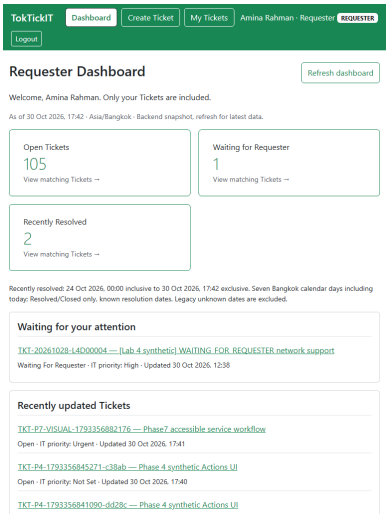
mobile full original

Comparison extracts only: inspect grid/stacking/controls, not tiny body text. Readable operational panels appear in Parts5-8; complete long originals are linked. These are real Phase7 local captures, not new main evidence.

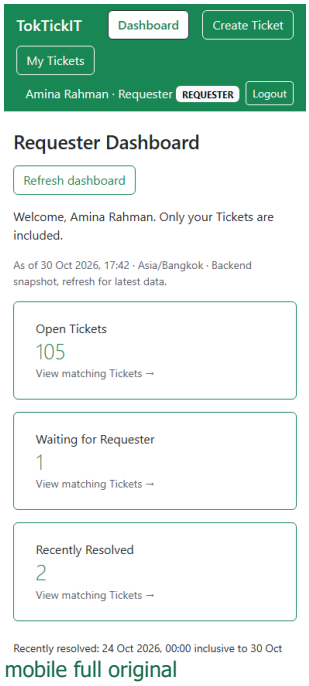
Requester Dashboard - three-width layout comparison



desktop full original



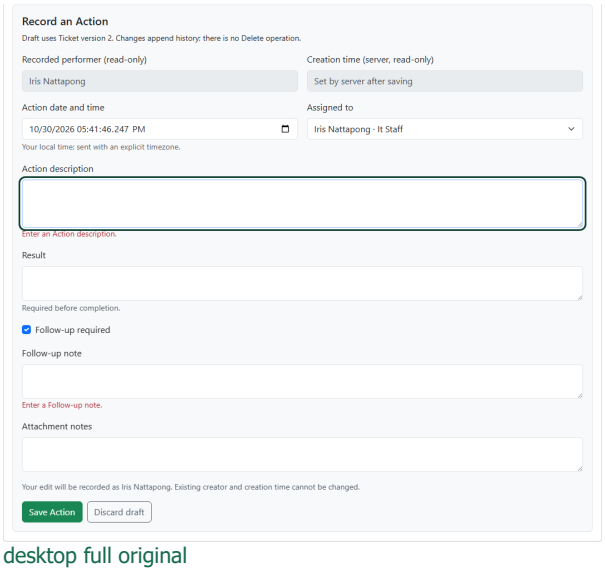
tablet full original



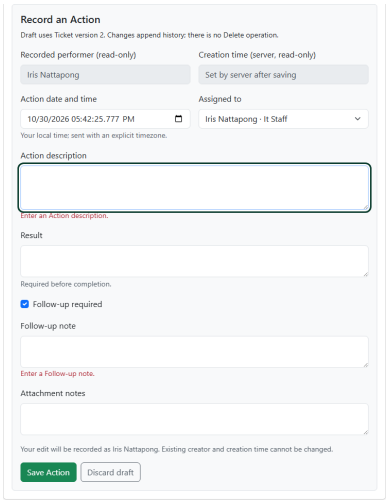
mobile full original

Comparison extracts only: inspect grid/stacking/controls, not tiny body text. Readable operational panels appear in Parts5-8; complete long originals are linked. These are real Phase7 local captures, not new main evidence.

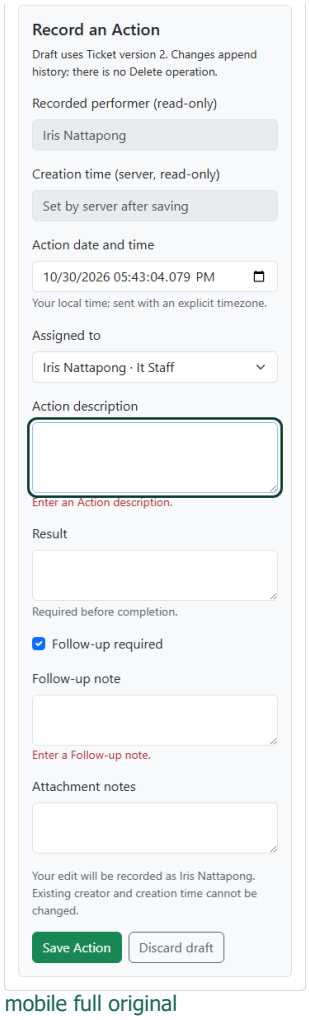
Actions Taken validation - three-width layout comparison



desktop full original



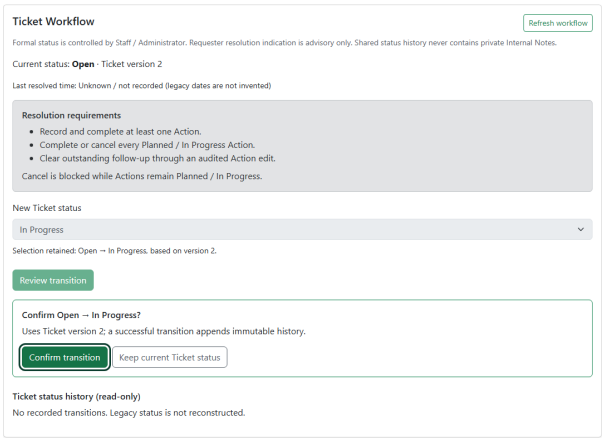
tablet full original



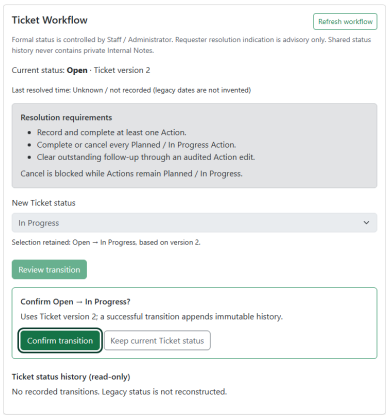
mobile full original

Comparison extracts only: inspect grid/stacking/controls, not tiny body text. Readable operational panels appear in Parts5-8; complete long originals are linked. These are real Phase7 local captures, not new main evidence.

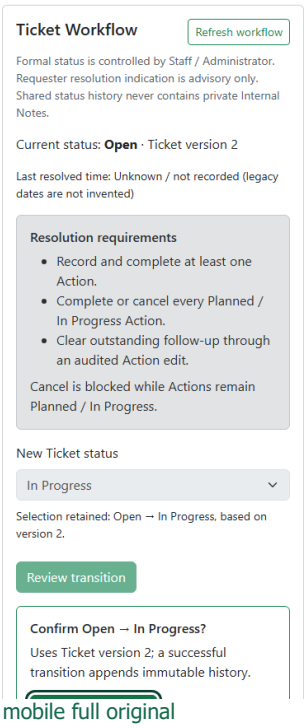
Ticket workflow confirmation - three-width layout comparison



desktop full original



tablet full original



mobile full original

Comparison extracts only: inspect grid/stacking/controls, not tiny body text. Readable operational panels appear in Parts5-8; complete long originals are linked. These are real Phase7 local captures, not new main evidence.